

NON-CARTESIAN INTERNAL CATEGORIES AND DISTRIBUTORS WITHOUT REGULARITY: A POLY-VIRTUAL APPROACH

ABSTRACT. Aguiar’s non-Cartesian internal categories use bicomodule cotensors for composition. We define formal internal categories in every monoidal category on actual bicomodule carriers, using a poly-virtual calculus with finite paths on both boundaries. A forest envelope and localization make intrinsic globular contextually universal comparisons reversible. Substitution presheaves and backward evaluation prove conservativity, including equivariant maps over changing comonoid bases. Stable cotensors recover the usual category structures, functors, and bimodule actions.

Formal categories then form the objects of an equipment of distributors on bicomodule-path carriers. Cartesian factorization constructs their restrictions; a second conservative completion supplies many-sided cells and coherent boundary transport. Contextually preserved action coequalizers represent distributor composition. Finite group-like bases in abelian groups recover enriched categories and distributors despite ambient nonregularity. A stronger triangular example has a nonzero action on one actual bicomodule whose input path has no contextual atomic representative: restriction along a formal functor preserves the action while destroying stability of its cotensor. The general completion also preserves virtual monoids and their generalized-multicategory examples.

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1. INTRODUCTION

An internal category has an object of objects, an object of arrows, and composition on composable pairs. In a monoidal category, Aguiar expresses these data by a comonoid C , a C – C bicomodule A , and maps

$$m : A \square_C A \longrightarrow A, \quad u : C \longrightarrow A,$$

with associative and unital multiplication [Agu97, Definition 2.3.1]. The comonoid and its coactions replace the object of objects and the source and target maps; the cotensor replaces the pullback of composable arrows. This gives a formulation that also makes sense when the monoidal product is not Cartesian.

Cotensor composition requires equalizers and their compatibility with tensoring. Aguiar’s regularity condition supplies this compatibility for every equalizer in the ambient monoidal category [Agu97, Definition 2.1.1]. However, even $(\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z})$ fails this condition. As [proposition 15.4](#) shows, the finite group-like comonoids in $\mathbf{Ab}$ nevertheless admit stable bicomodule composition and encode the expected finite-object enriched categories. A general definition should therefore separate the specification of category operations from the existence of all ambient horizontal composites.

Cruttwell and Shulman’s virtual framework provides a precedent for this separation. It treats enriched categories, internal categories, and generalized multicategories through cells with several horizontal inputs and one output [CS10]. In particular, its virtual equipment of enriched profunctors is available without the coends used to represent horizontal profunctor composition. The relevant analogy is between these coends and bicomodule cotensors. Coends here compose profunctors between enriched categories; the composition operations inside an enriched category are already part of its defining data.

The guiding question of this paper is therefore the following: can non-Cartesian internal categories be defined over an arbitrary monoidal category, on actual bicomodule carriers, so that stable cotensors recover Aguiar’s multiplication and changing-base functors, and can these formal categories themselves be the objects of an equipment of distributors? The primitive maps available without cotensors have the form

$$N \longrightarrow M_1 \otimes \cdots \otimes M_n,$$

subject to balance and equivariance equations. They describe cones into a prospective composite. Their arity is opposite to that of multiplication. Consequently, the existing virtual bicomodule calculus alone does not yet supply the desired forward category operations.

We construct a two-sided cell calculus in which both boundaries are finite paths of bicomodules. A formal internal category has an endobicomodule $A : C \rightharpoonup C$ and cells

$$\mu : (A, A) \Longrightarrow A, \quad \eta : \emptyset_C \Longrightarrow A \quad (1.1)$$

satisfying the monad equations. The construction makes each existing intrinsic globular contextually universal composite comparison reversible. It thereby allows represented cotensors to be used on either boundary, while paths without representatives remain legitimate boundaries. Its cells are specified by finite expressions and relations, so the definition does not depend on the existence of the object $A \square_C A$.

Theorem 1.1 (Internal categories without regularity). *Every monoidal category $\mathcal{V}$ determines a poly-virtual equipment $\mathbf{Q}(\mathcal{V})$ and a category $\mathbf{FCat}(\mathbf{Q}(\mathcal{V}))$ with the following properties.*

- (1) *The objects of $\mathbf{Q}(\mathcal{V})$ are comonoids, its vertical arrows are comonoid maps, and its designated atomic horizontal arrows are actual bicomodules. Its unary squares between designated atomic bicomodules are exactly equivariant bicomodule maps. More generally, all primitive balanced cells are preserved and reflected.*
- (2) *A formal internal category consists of an atomic endobicomodule and cells as in [equation \(1.1\)](#), with associativity and unit equations given by tree substitution. Homomorphisms have ordinary forward variance and may change the comonoid base.*
- (3) *On bicomodule paths represented by cotensor cones preserved under tensoring, these structures, their homomorphisms, and their bimodule actions recover the corresponding cotensor structures and equations. In particular, for regular $\mathcal{V}$ the category $\mathbf{FCat}(\mathbf{Q}(\mathcal{V}))$ is isomorphic to Aguiar’s category of internal categories and internal functors.*

The construction applies more generally to virtual equipments. Here **contextually universal** means that a comparison represents its source path in every compatible horizontal context, including contexts that pass through different objects. The technical result is a conservative completion that recognizes all such comparisons intrinsically.

Theorem 1.2 (Conservative completion). *Every virtual equipment $\mathbb{D}$ has a merge realization $\mathbf{E}(\mathbb{D})$ and a poly-virtual reduct $U_{\text{poly}}\mathbf{E}(\mathbb{D})$ on the same objects, vertical arrows, and designated atomic horizontal arrows, with natural bijections*

$$\mathbb{D}_{f,g}(\Gamma; H) \cong \mathbf{E}(\mathbb{D})_{f,g}(\Gamma; H)$$

for all original paths Γ , atomic outputs H , and compatible vertical boundaries. The realization makes every intrinsic globular contextually universal comparison, and hence every horizontal contextual instance of one, reversible. It also admits coherent transport of horizontal boundaries. The reduct retains all cells and equalities and the tree operations; it preserves and reflects formal categories and their homomorphisms on atomic carriers.

The bicomodule instance is

$$\mathbf{Q}(\mathcal{V}) = U_{\text{poly}}(\mathbf{E}(\text{Mod}_{\text{CS}}(\mathcal{V}^{\text{op}}))^{\vee}).$$

Here Mod_{CS} is Leinster’s monoids-and-bimodules construction [Lei04, Section 5.3], in the virtual-double-category formulation of Cruttwell and Shulman [CS10, Definition 2.8]. It supplies the initial balanced calculus; the completion and the reversal of squares supply its two-sided, forward formulation.

The distributor construction uses the same completion a second time. Let $\mathbb{B}(\mathcal{V}) = \mathbf{E}(\mathbf{C}(\mathcal{V}))^{\vee}$ be the merge realization, and regard its finite bicomodule paths as individual horizontal arrows of

an underlying virtual equipment $\mathbb{K}(\mathcal{V})$. We form $\text{Mod}_{\text{CS}}(\mathbb{K}(\mathcal{V}))$ and retain exactly the monoids whose carriers are original atomic bicomodules. The restriction here is on objects: the horizontal carriers of the resulting equipment $\mathbb{D}_{\text{dist}}(\mathcal{V})$ remain finite bicomodule paths. This makes the required distributor restrictions available while retaining all original atomic bimodules. Define

$$\text{Dist}_{\text{poly}}(\mathcal{V}) = U_{\text{poly}}\mathbf{E}(\mathbb{D}_{\text{dist}}(\mathcal{V})).$$

Theorem 1.3 (Formal distributors without regularity). *For every monoidal category $\mathcal{V}$, $\text{Dist}_{\text{poly}}(\mathcal{V})$ is a poly-virtual equipment with the following properties.*

- (1) *Its objects are the formal internal categories in $\mathbf{Q}(\mathcal{V})$, and its vertical category is exactly $\mathbf{FCat}(\mathbf{Q}(\mathcal{V}))$.*
- (2) *Its designated horizontal arrows are distributors with associative, unital, commuting actions on finite bicomodule-path carriers. On a carrier consisting of one actual bicomodule, these are exactly the atomic bimodule actions in $\mathbf{Q}(\mathcal{V})$.*
- (3) *Balanced single-output distributor cells are preserved and reflected by the completion, including all their changing-base instances. In particular, unary cells between distributors carried by one actual bicomodule are exactly equivariant bicomodule maps preserving both actions.*
- (4) *Formal functors have companions and conjoiners, yielding coherent transport of distributor boundaries. Every intrinsic globular contextually universal distributor-composition comparison becomes reversible.*

There are two distinct representation problems. Cotensors represent paths of bicomodules over comonoid bases. Relative tensor products of distributors represent balancing over the actions of an intervening formal category. We give a contextual coequalizer construction for the latter and verify it in the finite group-like **Ab** sector. Neither construction requires a missing representative to be an object of $\mathcal{V}$.

The distinction occurs inside an action in [theorem 16.9](#). There, a formal category with carrier $A \cong \mathbb{Z}^5$ acts on an actual bicomodule $M \cong (\mathbb{Z}/4)^2$, although (A, M) has no contextual atomic representative. Both categories used to construct the action have stable finite composition paths. Restriction along a formal functor produces the action from a stable one, showing that stability of action cotensors need not survive restriction, even over an identity comonoid map.

The term **poly-virtual equipment** has the precise meaning given in [definitions 3.3](#) and [7.6](#). Its equipment structure transports boundaries, and a transported boundary can itself be a path. The tree and merge operations are related to those of [\[Had19\]](#); empty boundaries and a vertical category are included here. The relation with implicit structures is discussed in [section 18](#).

The construction and conservativity occupy [sections 2 to 8](#); [sections 9 to 11](#) give the bicomodule application and cotensor recovery. Distributors and their representations are developed in [sections 12 to 14](#). The examples in [sections 15](#) and [16](#) separate ambient nonregularity from failure of representation at an action boundary. Further applications and comparison with other envelopes follow in [sections 17](#) and [18](#).

Conventions. Horizontal paths are written in traversal order: HK means first H , then K . Composition of ordinary maps and of cells is written in functional order: $ba = b \circ a$. Horizontal juxtaposition of cells is $a \odot b$. The path of length zero at X is $\emptyset_X$; a represented unit is denoted U_X . A semicolon separates the two boundaries of a cell. Associators and unitors in a monoidal category are inserted by their canonical coherence maps; no braiding is used unless explicitly stated. The notation $\mathcal{V}^{\text{op}}$ reverses maps and leaves the order of tensor factors unchanged. Size is discussed in [section C](#). We write $\mathbf{A} = (X, A, \mu_A, \eta_A)$ for a formal category and $\mathbf{F} = (f, F)$ for a formal functor, separating the whole structure from its carrier and its underlying square. The word **atomic** is relative to the designated horizontal graph: an atomic arrow of

the distributor equipment is one distributor, whose carrier in the original bicomodule graph can be a path. We use the slashed proarrow $\rightharpoonup$ of [CS10, Definition 2.1]; its slash is a proarrow marker. Double-category diagrams use slashed horizontal arrows and a centered double arrow for the cell. In diagrams whose vertices are paths, ordinary arrows denote cells in the merge realization. Commutativity asserts equality of composites; the category and action laws compare tree composites. Dashed arrows denote unique factorizations, except where explicitly labelled $\#$.

Note on use of AI. Generative AI systems were used during the research and development of this manuscript. Responsibility for the mathematical content, arguments, and conclusions rests with the author.

2. BICOMODULES AND THE MISSING COMPOSITION

2.1. Virtual composition and its enriched precedent. A virtual double category has objects, vertical maps, horizontal arrows, and cells from a finite path to one horizontal arrow. Its composition is substitution of compatible cells. A cell is globular when its exterior vertical maps are identities. A composite of a horizontal path, when it exists, represents these substitutions in every surrounding path context. A virtual equipment has represented horizontal units and cartesian restrictions along pairs of vertical arrows [CS10, Sections 2, 5, and 7].

For enriched profunctors $P : \mathcal{A} \rightharpoonup \mathcal{B}$ and $R : \mathcal{B} \rightharpoonup \mathcal{C}$, the represented composite has components

$$(P \odot R)(a, c) = \int^{b \in \mathcal{B}} P(a, b) \otimes R(b, c).$$

Balanced families of maps out of these tensor factors define the virtual cells before the coend is formed. This is the distinction between the virtual equipment and its represented composites [CS10, Examples 5.6 and 7.7]. For bicomodules the basic balanced maps point into tensor products. We now specify those maps without taking their equalizers.

2.2. Comonoids, coactions, and balanced cells.

Definition 2.1. A comonoid in $\mathcal{V}$ is an object C with coassociative $\delta_C : C \rightarrow C \otimes C$ and counit $\varepsilon_C : C \rightarrow I$. A comonoid map preserves both maps. A bicomodule $M : C \rightharpoonup D$ has coactions

$$\lambda_M : M \rightarrow C \otimes M, \quad \rho_M : M \rightarrow M \otimes D$$

which are counital and coassociative and commute:

$$\begin{array}{ccc} M & \xrightarrow{\lambda_M} & C \otimes M \\ \rho_M \downarrow & & \downarrow 1_C \otimes \rho_M \\ M \otimes D & \xrightarrow{\lambda_M \otimes 1_D} & C \otimes M \otimes D. \end{array} \quad (2.1)$$

A map $F : M \rightarrow N$ over comonoid maps $f : C \rightarrow C'$, $g : D \rightarrow D'$ is equivariant if

$$\begin{array}{ccc} M & \xrightarrow{F} & N \\ \lambda_M \downarrow & & \downarrow \lambda_N \\ C \otimes M & \xrightarrow{f \otimes F} & C' \otimes N \end{array} \quad \begin{array}{ccc} M & \xrightarrow{F} & N \\ \rho_M \downarrow & & \downarrow \rho_N \\ M \otimes D & \xrightarrow{F \otimes g} & N \otimes D'. \end{array} \quad (2.2)$$

Our horizontal arrow points from the left coaction base to the right coaction base. References using the opposite horizontal direction write composite bicomodules in the opposite order. This convention fixes the order of all tensor factors below.

For $M : C \rightharpoonup D$ and $N : D \rightharpoonup E$, the prospective cotensor is the equalizer of

$$\rho_M \otimes 1_N, 1_M \otimes \lambda_N : M \otimes N \rightrightarrows M \otimes D \otimes N.$$

Even if this equalizer is unavailable, maps equalizing the displayed pair can still be specified. Longer paths impose one such balance equation at each interior base, together with the exterior coaction equations.

Definition 2.2 (Balanced cells). Let $M_i : C_{i-1} \rightharpoonup C_i$ for $1 \leq i \leq n$, let $N : D \rightharpoonup E$, and let $a : D \rightarrow C_0, b : E \rightarrow C_n$ be comonoid maps. For $n > 0$, set $T = M_1 \otimes \cdots \otimes M_n$. A balanced cell is a map $\phi : N \rightarrow T$ with exterior equivariance

$$\begin{array}{ccc} N & \xrightarrow{\phi} & T \\ \lambda_N \downarrow & & \downarrow \lambda_{M_1} \otimes 1 \\ D \otimes N & \xrightarrow{a \otimes \phi} & C_0 \otimes T \end{array} \quad \begin{array}{ccc} N & \xrightarrow{\phi} & T \\ \rho_N \downarrow & & \downarrow 1 \otimes \rho_{M_n} \\ N \otimes E & \xrightarrow{\phi \otimes b} & T \otimes C_n \end{array} \quad (2.3)$$

and one interior balance at each C_i . Write T_i for T with C_i inserted between M_i and M_{i+1} , and let $r_i = 1 \otimes \rho_{M_i} \otimes 1$ and $\ell_i = 1 \otimes \lambda_{M_{i+1}} \otimes 1$. Balance says that the two routes have a common composite ψ_i :

$$\begin{array}{ccc} N & \xrightarrow{\phi} & T \\ & \searrow \psi_i & \downarrow \ell_i \downarrow r_i \\ & & T_i. \end{array} \quad (1 \leq i < n). \quad (2.4)$$

For $n = 0$, the path is $\emptyset_C$, $a : D \rightarrow C, b : E \rightarrow C$, and a balanced cell is a map $\phi : N \rightarrow I$ making

$$\begin{array}{ccc} N & \xrightarrow{\rho_N} & N \otimes E \\ \lambda_N \downarrow & & \downarrow \phi \otimes b \\ D \otimes N & \xrightarrow{a \otimes \phi} & C \end{array} \quad (2.5)$$

commute, with unitors understood.

2.3. The virtual equipment already available. The balanced maps are closed under substitution. Their orientation as single-output cells is forced by that substitution law: the map from N into a tensor of M_i is regarded as a cell with output N . This produces the following instance of the monoids-and-bimodules construction of Leinster, in the virtual-double-category formulation of Cruttwell and Shulman [Lei04, Section 5.3] [CS10, Definition 2.8].

Proposition 2.3. *Comonoids, opposite comonoid maps, bicomodules, and the balanced cells of definition 2.2, regarded as cells $(M_1, \dots, M_n) \Longrightarrow N$, form the virtual equipment*

$$\mathbb{C}(\mathcal{V}) = \text{Mod}_{\text{CS}}(\mathcal{V}^{\text{op}}).$$

This construction exists for every monoidal $\mathcal{V}$.

The balanced map $\phi : N \rightarrow M_1 \otimes \cdots \otimes M_n$ has profile

$$\begin{array}{ccc} C_0 & \xrightarrow{(M_1, \dots, M_n)} & C_n \\ a^{\text{op}} \downarrow & \Downarrow \phi & \downarrow b^{\text{op}} \\ D & \xrightarrow{N} & E \end{array}$$

in the opposite comonoid category.

Proof.

- (1) **Identify the module construction.** Regard $\mathcal{V}^{\text{op}}$ as the one-object virtual double category whose cells are maps into tensor products in $\mathcal{V}$. The construction $\text{Mod}_{\mathcal{C}\mathcal{S}}$ requires no additional exactness condition on this virtual double category [Lei04, Section 5.3][CS10, Remark following Definition 2.8]. A monoid there is precisely a comonoid in $\mathcal{V}$, and a bimodule is precisely a bicomodule. The exterior action equations and interior balancing equations in the module construction become equations (2.3) and (2.4); the zero-arity action equation becomes equation (2.5).
- (2) **Check substitution at nonempty blocks.** Explicitly, substitution composes ϕ with the tensor product of the maps substituted into its factors. An interior equation inside a block is the corresponding equation of the substituted map, tensored with the other maps. At an interface between two nonempty blocks, equation (2.3) moves the coactions through those maps and the remaining equality is equation (2.4) for ϕ .
- (3) **Handle empty blocks.** An empty block is removed using its map to I ; equation (2.5) identifies the two induced maps to its vertex comonoid, so the same interface equality remains true. Repeating this for successive empty blocks proves the statement also when every block is empty.
- (4) **Verify the exterior and substitution laws.** The exterior equations follow by the first and last such calculations. Associativity and identities now follow from tensor functoriality and composition in $\mathcal{V}$.
- (5) **Construct restrictions.** Restrictions are explicit. To restrict $N : D \nrightarrow E$ along vertical arrows $C \rightarrow D$ and $C' \rightarrow E$ in the opposite comonoid category, write their actual maps as $a : D \rightarrow C$, $b : E \rightarrow C'$. Retain the object N and replace its coactions by $(a \otimes 1)\lambda_N$ and $(1 \otimes b)\rho_N$.
- (6) **Verify cartesianness.** The identity map on N is the restriction cell. The equations for factorization through it are literally equation (2.3) with these coactions, proving cartesianness.
- (7) **Supply units.** Units are proved in lemma 2.4 below. This also directly verifies the units and restrictions supplied abstractly by [CS10, Propositions 5.5 and 7.4]. $\square$

Lemma 2.4 (The nullary representative). *For a comonoid C , its regular bicomodule $(C, \delta_C, \varepsilon_C)$ represents $\emptyset_C$ in $\mathbb{C}(\mathcal{V})$. The universal cell $\emptyset_C \Rightarrow C$ is represented by ε_C .*

Proof.

- (1) **Construct the nullary factor.** First consider a globular nullary cell $\phi : N \rightarrow I$. By equation (2.5),

$$q = (1_C \otimes \phi)\lambda_N = (\phi \otimes 1_C)\rho_N : N \rightarrow C.$$
- (2) **Verify equivariance.** Coassociativity of λ_N gives $\delta_C q = (1_C \otimes q)\lambda_N$, and that of ρ_N gives $\delta_C q = (q \otimes 1_C)\rho_N$; in each equation use the corresponding expression for q . Thus q is a bicomodule map.
- (3) **Prove nullary uniqueness.** Counitality gives the factorization triangle

$$\begin{array}{ccc}
 N & \xrightarrow{\exists! q} & C \\
 & \searrow \phi & \downarrow \varepsilon_C \\
 & & I.
 \end{array}$$

Conversely a bicomodule map $q : N \rightarrow C$ satisfies both displayed expressions with $\phi = \varepsilon_C q$, proving uniqueness.

- (4) **Insert the unit in a context.** For a contextual instance, take a balanced map $\phi : N \rightarrow \otimes L \otimes \otimes R$ in which the two paths meet at C . If L is nonempty, insert C by the right coaction of its last factor. If R is nonempty, one can instead use the left coaction of its first factor. Interior balance says these two constructions agree when both apply. If both are empty, use the map q just constructed. If only one side is present, the exterior equation supplies the missing endpoint calculation.
- (5) **Check the contextual equations.** Coassociativity gives balance at the two new interfaces, and commuting coactions give the unchanged exterior equations.
- (6) **Prove contextual uniqueness.** Removing the inserted factor by ε_C recovers ϕ . Conversely, either balance equation adjacent to an inserted C , followed by its counit, forces this insertion formula. Thus the factorization is unique in all contexts.
- (7) **Allow changing exterior maps.** For nonidentity exterior maps, corestrict the two exterior coactions by the explicit restriction construction above. The resulting factorization is the globular one just proved, with the same underlying maps in $\mathcal{V}$. $\square$

3. TWO-SIDED CELLS AND TREE PASTINGS

Definition 3.1 (Boundary signature). A boundary signature consists of a category $\mathcal{C}$ and a directed graph of horizontal arrows on its objects. A horizontal path $\Gamma = (H_1, \dots, H_n) : X_0 \rightarrowtail X_n$ includes its entire sequence of vertices. A cell profile is $(f, g; \Gamma; \Delta)$, where $\Gamma : X \rightarrowtail Y$, $\Delta : X' \rightarrowtail Y'$, $f : X \rightarrow X'$, and $g : Y \rightarrow Y'$ lie in $\mathcal{C}$. We depict it as

$$\begin{array}{ccc}
 X & \xrightarrow{\Gamma} & Y \\
 f \downarrow & \Downarrow \alpha & \downarrow g \\
 X' & \xrightarrow{\Delta} & Y'
 \end{array}$$

A globular profile has $f = \text{id}_X$ and $g = \text{id}_Y$.

We specify the pasting shapes independently of any cells that will label them. Start with rectangles with finitely many ordered horizontal ports on their upper and lower sides. The horizontal ports are labelled by arrows and their vertices; the two lateral sides are labelled by composable vertical paths. A rectangular scheme is a finite expression formed by horizontal and vertical pasting of such rectangles, identity strips on horizontal paths, and empty horizontal strips on vertical arrows. Pasting requires literal agreement of the full common boundary. In particular, horizontal pasting requires equality of the common vertical arrow, not merely agreement of its endpoints. Empty strips compose as the arrows of $\mathcal{C}$ do.

The **wire graph** of a scheme is the finite multigraph whose vertices are its labelled rectangles and whose edges are its internal atomic horizontal wires; structural identity strips are contracted. Acyclicity is understood for this multigraph, so in particular two parallel edges form a cycle. A **tree scheme** is a rectangular scheme whose wire graph is nonempty, connected, and acyclic, and which has no separate horizontal wire component meeting no labelled rectangle. The last condition excludes adjoining unused through-wires as a tree operation. For the poly-virtual calculus the permitted schemes with no labelled rectangle are a single identity wire and an empty horizontal square on a vertical arrow. Call these **elementary structural schemes**. Two schemes describe the same pasting when they differ by planar deformation, insertion or deletion of structural identity strips, associativity of pasting, or interchange of rectangular subdivisions. The arity of an internal seam is its number of atomic wires. Thus two boxes joined along two wires are a merge scheme, not a tree scheme.

Lemma 3.2. *Substitution of a tree scheme into a labelled rectangle of another tree scheme again gives a tree scheme. Insertion of an elementary structural scheme gives a tree scheme or an elementary structural scheme. Such substitution is associative and has one-rectangle schemes as identities. Evaluation orders related by contracting connected rectangular subtrees describe the same substituted scheme.*

Proof.

- (1) **Preserve the tree shape.** Rectangular substitution preserves every boundary label and replaces one graph vertex by a tree. Each old incident edge attaches to its designated port. The resulting graph is connected. A cycle entirely in the inserted graph is impossible; any other cycle would contract to a cycle in the old graph.
- (2) **Prove substitution associativity.** Substitution of finite labelled rectangles is associative before passing to planar equivalence. The equivalence respects substitution since each of its generating moves can be performed inside the substituted rectangle.
- (3) **Check structural substitutions.** A single identity wire removes a vertex having one input and one output, preserving connectivity and acyclicity. An empty structural square can replace only a zero-port vertex, which in a connected scheme must be its sole vertex. These observations prove the structural cases.
- (4) **Obtain the evaluation identity.** The last assertion is the resulting associativity identity. $\square$

Definition 3.3 (Poly-virtual and merge calculi). A **poly-virtual double calculus** on a boundary signature assigns a collection $\mathbb{P}_{f,g}(\Gamma; \Delta)$ to every profile and evaluates every labelled tree scheme. It supplies the cells for single identity wires and structural empty squares, and evaluates elementary structural schemes. Evaluation respects the equivalences of schemes and substitution of schemes into rectangles, and evaluation of a one-rectangle scheme is its label. A map preserves this data.

A **merge calculus** has evaluation of all rectangular schemes, with the same equations, including all structural schemes on arbitrary paths. Its **poly-virtual reduct** $U_{\text{poly}}\mathbb{B}$ keeps the same cell collections and equalities and restricts the specified evaluation operations to tree and elementary structural schemes. In particular an identity cell on a longer path remains a cell, but its empty multiwire scheme is not a designated poly-virtual operation. Original horizontal arrows are called **atomic** when a calculus is presented using their paths.

This is an algebraic definition: lemma 3.2 constructs the multicategory of allowed operations, and a calculus is an algebra for those operations. It does not require every many-sided pasting to be evaluable.

For clarity, an elementary globular cut can be described without pictures. Let

$$\alpha : \Gamma \Longrightarrow (\Delta_0, M, \Delta_1), \quad \beta : (\Lambda_0, M, \Lambda_1) \Longrightarrow \Sigma.$$

The planar two-rectangle cut is allowed when all endpoints match and

$$(\Delta_0 = \emptyset \text{ or } \Lambda_0 = \emptyset), \quad (\Delta_1 = \emptyset \text{ or } \Lambda_1 = \emptyset). \quad (3.1)$$

Its profile is

$$(\Lambda_0, \Gamma, \Lambda_1) \Longrightarrow (\Delta_0, \Sigma, \Delta_1).$$

Indeed, these conditions say that on either side of the common wire only one box has an exposed boundary to traverse. If both corresponding contexts are nonempty, the putative exterior is not the stated pair of directed paths. One useful instance is

$$\frac{\Gamma \Longrightarrow (\Delta, M) \quad (M, \Lambda) \Longrightarrow \Sigma}{(\Gamma, \Lambda) \Longrightarrow (\Delta, \Sigma)}. \quad (3.2)$$

Proposition 3.4. *The single-output fragment of a poly-virtual calculus is a virtual double category. Its single-input fragment is the vertically reversed version of a virtual double category. Ordinary virtual substitution, including its changing-base and nullary instances, is an allowed tree operation.*

Proof.

- (1) **Identify the substitution tree.** An outer single-output rectangle and a compatible family of single-output rectangles above its input ports form a rooted tree. Shared lateral maps are precisely the compatibility data of virtual substitution.
- (2) **Verify the virtual laws.** Replacing vertices of this tree gives associativity by [lemma 3.2](#); structural strips give the identity laws. A nullary vertex has no incoming wire and causes no change in this argument.
- (3) **Reverse the boundaries.** Reversing the upper and lower boundaries proves the single-input assertion.
- (4) **Match the whole family.** Vertical maps must be matched for the whole family; they need not allow the family to be pasted one member at a time with other members held fixed. $\square$

Remark 3.5 (Empty profiles). A cell with two empty boundaries has a specified upper object, a specified lower object, and two lateral arrows between them. It is an ordinary possible label of a zero-port rectangle. Its arbitrary vertical composites are available in a merge calculus; they are not automatically tree operations. Structural empty squares on a vertical arrow are distinguished and obey the category laws. These conventions include monad units without identifying an empty boundary with an atomic horizontal unit.

4. FORMAL CATEGORIES AND THEIR HOMOMORPHISMS

Definition 4.1. A **formal category** in $\mathbb{P}$ is a quadruple (X, A, μ, η) , where $A : X \rightharpoonup X$ is an atomic arrow and $\mu : (A, A) \Longrightarrow A$, $\eta : \emptyset_X \Longrightarrow A$ are globular cells such that

$$\begin{array}{ccc}
 (A, A, A) & \xrightarrow{(\mu, 1_A)} & (A, A) \\
 (1_A, \mu) \downarrow & & \downarrow \mu \\
 (A, A) & \xrightarrow{\mu} & A
 \end{array}
 \qquad
 \begin{array}{ccc}
 (A, A) & \xleftarrow{(\eta, 1_A)} & A \xrightarrow{(1_A, \eta)} (A, A) \\
 \mu \searrow & & \parallel \\
 & & A. \\
 \mu \swarrow & & \swarrow \mu
 \end{array}
 \tag{4.1}$$

Each expression denotes the corresponding rooted tree evaluation. A homomorphism $(X, A) \rightarrow (Y, B)$ is a vertical arrow $f : X \rightarrow Y$ and a cell $F \in \mathbb{P}_{f,f}(A; B)$ satisfying

$$\begin{array}{ccc}
 (A, A) & \xrightarrow{\mu_A} & A \\
 (F, F) \downarrow & & \downarrow F \\
 (B, B) & \xrightarrow{\mu_B} & B
 \end{array}
 \qquad
 \begin{array}{ccc}
 \emptyset_X & \xrightarrow{\eta_A} & A \\
 [f] \downarrow & & \downarrow F \\
 \emptyset_Y & \xrightarrow{\eta_B} & B.
 \end{array}
 \tag{4.2}$$

Here $[f]$ is the structural empty square on f ; the notation $\eta_B[f]$ denotes its paste above η_B .

Each complete composite in [equations \(4.1\)](#) and [\(4.2\)](#) is a tree evaluation. Individual whiskerings, such as $(\mu, 1_A)$, are displayed in the merge realization; adjoining an unused identity wire is not itself a designated tree operation.

Proposition 4.2. *Formal categories and their homomorphisms form a category $\mathbf{FCat}(\mathbb{P})$.*

Proof.

- (1) **Define composition.** Compose (f, F) and (g, G) by (gf, GF) , using unary tree pasting. Associativity and identities follow from the corresponding tree equations.
- (2) **Preserve multiplication.** For multiplication, the homomorphism equations for F and G , interleaved with substitution associativity, give

$$\begin{aligned} (GF)\mu_A &= G(F\mu_A) = G(\mu_B(F, F)) \\ &= (G\mu_B)(F, F) = \mu_C(G, G)(F, F) = \mu_C(GF, GF). \end{aligned}$$

- (3) **Preserve units.** For units, the structural empty squares compose with their vertical arrows, so $(GF)\eta_A = G\eta_B[f] = \eta_C[g][f] = \eta_C[gf]$. Thus the composite satisfies both homomorphism equations. The same equations hold for identity cells by the substitution identity laws. $\square$

Theorem 4.3 (Invariance under the reduct). *For every merge calculus with designated atomic carriers, the identity on data induces an isomorphism of categories*

$$\mathbf{FCat}(\mathbb{B}) \cong \mathbf{FCat}(U_{\text{poly}}\mathbb{B}).$$

Proof.

- (1) **Retain the defining cells.** All cells specifying an object or morphism occur in the retained cell collections.
- (2) **Retain the defining equations.** Every operation in [equations \(4.1\)](#) and [\(4.2\)](#) is a tree or structural operation. The reduct retains their values and the equalities between them.
- (3) **Retain category composition.** Finally, identity and composite homomorphisms use only structural and unary tree operations. Consequently the object sets, hom-sets, identities, and compositions agree. $\square$

5. THE TYPED FOREST ENVELOPE

Let P be a globular virtual calculus: its objects form a set or collection, its arrows form a directed graph, and $P(\Gamma; H)$ has identities and substitution for composable paths with the same endpoints. In the main application P is the entire globular part of $\mathbb{D}$. Arrows between different objects are retained.

Definition 5.1. The strict path 2-category $\mathbf{For}(P)$ has the objects of P and its finite paths as 1-arrows. For paths $\Gamma = (H_1, \dots, H_n)$ and $\Delta = (K_1, \dots, K_m)$ with common endpoints, a 2-cell consists of a weakly increasing subdivision

$$0 = i_0 \leq i_1 \leq \dots \leq i_m = n$$

whose cut vertices are the vertices of Δ , and cells $a_j \in P(H_{i_{j-1}+1}, \dots, H_{i_j}; K_j)$. Repeated indices are permitted only with the indicated coincident vertices. When $m = 0$, there is a cell only if $n = 0$, and it is the empty identity forest. Vertical composition substitutes the a_j ; horizontal composition concatenates subdivisions and cells.

Lemma 5.2. *These operations make $\mathbf{For}(P)$ a strict 2-category, and*

$$\mathbf{For}(P)(\Gamma; H) = P(\Gamma; H) \tag{5.1}$$

for atomic H . A map from P to the single-output part of a strict path 2-category extends uniquely to a strict 2-functor from $\mathbf{For}(P)$.

Proof.

- (1) **Prove associativity.** Associativity of substitution proves vertical associativity; concatenation proves horizontal associativity.

- (2) **Prove interchange.** Substitution on disjoint consecutive blocks commutes, which is interchange.
- (3) **Verify the identity laws.** The forest of unary identities and the empty forest give the identity laws.
- (4) **Recover single-output cells.** With one output the subdivision has only one block, proving [equation \(5.1\)](#).
- (5) **Prove the extension property.** A forest must be sent to the horizontal composite of the images of its roots, so an extension is unique, and the same substitution identities show that it is a functor. $\square$

Definition 5.3 (Intrinsic comparison class). A cell $s \in P(\Omega; T)$ is **contextually universal** if for every compatible pair of paths L, R and every atomic output B , each $\alpha \in P(L, \Omega, R; B)$ factors uniquely as

$$\begin{array}{ccc}
 (L, \Omega, R) & \xrightarrow{1_L \odot s \odot 1_R} & (L, T, R) \\
 & \searrow \alpha & \downarrow \exists! \bar{\alpha} \\
 & & B.
 \end{array} \tag{5.2}$$

Equivalently, substitution gives a bijection $s^* : P(L, T, R; B) \rightarrow P(L, \Omega, R; B)$. Let $S(P)$ be the class of all such cells, including those with empty source. We say that Ω is **represented by** T when such an s exists.

Lemma 5.4. *Identities are contextually universal. Substitution of contextually universal comparisons into a contextually universal comparison is contextually universal. Contextual instances of such comparisons are stable under further adjoining contexts.*

Proof.

- (1) **Include identities.** The first assertion is the identity law.
- (2) **Close under substitution.** For the second, the substitution map of the resulting cell is the composite of the substitution maps for the outer comparison and its inputs, in their specified contexts. These maps are bijections.
- (3) **Enlarge the context.** For the last assertion, enlarge L, R in [equation \(5.2\)](#). This last statement concerns instances in the forest envelope, which may have several outputs; it does not change the definition of $S(P)$ as single-output cells. $\square$

Lemma 5.5 (Globular detection in an equipment). *If P is the globular part of a virtual equipment $\mathbb{D}$, then [equation \(5.2\)](#) is equivalent to the same factorization property with arbitrary exterior vertical boundaries in $\mathbb{D}$.*

Proof.

- (1) **Restrict the target.** For a target B and exterior arrows f, g , its cartesian restriction gives

$$\mathbb{D}_{f,g}(\Xi; B) \cong P(\Xi; B[f, g]).$$

- (2) **Transport the factorization test.** This bijection commutes with substitution in Ξ . Apply it to both sides of the desired factorization map and use [equation \(5.2\)](#) with output $B[f, g]$.
- (3) **Recover the globular case.** The converse takes identity vertical boundaries. $\square$

6. CONSERVATIVE LOCALIZATION

Definition 6.1. For each $s : \Omega \Longrightarrow T$ in $S(P)$, adjoin a formal 2-cell $s^{-1} : T \Longrightarrow \Omega$ to $\text{For}(P)$. Form the strict 2-category presented by these generators, the forest equations, and

$$s^{-1}s = 1_\Omega, \quad ss^{-1} = 1_T. \quad (6.1)$$

Denote it by $L(P)$. Equations are closed under horizontal and vertical composition. Equivalently, localize the hom-categories of $\text{For}(P)$ at all contextual instances $1_L \odot s \odot 1_R$, and extend horizontal composition to those localizations.

The equivalence in the definition follows directly from the universal property of category localization: horizontal composition sends an instance to another instance, so it extends uniquely, and its associativity and interchange follow by uniqueness. In particular, the construction does not merely invert comparisons in isolated hom-categories and then postulate compatibility with contexts.

Lemma 6.2 (Substitution presheaves). *For an atomic arrow $A : X \multimap Y$, the assignment $\Phi_A(\Gamma) = P(\Gamma; A)$ is a presheaf on $\text{For}(P)(X, Y)$. It takes every contextual comparison to a bijection and therefore extends uniquely to a presheaf $\bar{\Phi}_A$ on $L(P)(X, Y)$. The map $P(\Gamma; A) \rightarrow L(P)(\Gamma; A)$ is injective.*

Proof.

- (1) **Define the forest action.** The action of a forest is simultaneous substitution into its consecutive blocks. Identity and composition follow from the virtual identities.
- (2) **Extend across inverses.** The assertion about comparisons is precisely [equation \(5.2\)](#). Thus the action of a formal inverse is the inverse bijection, and this defines the extension.
- (3) **Separate original cells.** If original cells a, b have equal images, apply their presheaf actions to $1_A \in P(A; A)$. The results are a and b , respectively, so $a = b$. $\square$

Lemma 6.3 (Backward evaluation). *Every $z \in L(P)(\Gamma; A)$ with atomic output A equals the image of the original cell $\bar{\Phi}_A(z)(1_A)$.*

Proof.

- (1) **Choose a layered expression.** Write z as a finite vertical sequence of forests and inverses of contextual comparisons. Such a sequence exists: distribute horizontal composites of generators into vertical layers by interchange, adding identity forests.
- (2) **Initialize backward evaluation.** Start at the final boundary A with 1_A and traverse the sequence backwards.
- (3) **Pass an ordinary forest.** At an ordinary forest $a : \Xi \Longrightarrow \Psi$, replace the current $b \in P(\Psi; A)$ by its substitution ba .
- (4) **Pass an inverse comparison.** At the inverse of

$$t = 1_L \odot s \odot 1_R : (L, \Omega, R) \Longrightarrow (L, T, R),$$

the current cell has source (L, Ω, R) . Let c be its unique factor through t using [equation \(5.2\)](#), so that $b = ct$. Then in the localization $bt^{-1} = c$.

- (5) **Complete the induction.** In either case the original cell now obtained equals the localized suffix that it replaces. Induction over the finite sequence proves the assertion.
- (6) **Prove independence of expression.** These two backward operations are exactly the actions defining $\bar{\Phi}_A$, which also proves independence from the sequence used. $\square$

Theorem 6.4 (Conservativity). *The canonical comparison is a bijection*

$$P(\Gamma; A) \xrightarrow{\cong} L(P)(\Gamma; A) \quad (6.2)$$

for every original path and atomic output. It respects all original substitutions. Reversing cells gives the corresponding single-input statement for the dual construction.

Proof.

- (1) **Establish the bijection.** Injectivity is [lemma 6.2](#) and surjectivity is [lemma 6.3](#).
- (2) **Preserve substitution.** The canonical map respects substitution by its definition.
- (3) **Obtain the dual assertion.** Reversing the displayed bijections proves the dual assertion. $\square$

Theorem 6.5 (Detection of represented paths). *For an original path Γ and atomic arrow T ,*

$$\Gamma \cong T \text{ in } L(P) \iff \Gamma \text{ is represented by } T \text{ in } P.$$

More precisely, an original cell $s : \Gamma \Rightarrow T$ is invertible in $L(P)$ if and only if $s \in S(P)$.

Proof.

- (1) **Invert universal comparisons.** The reverse implication is the construction.
- (2) **Recover an original comparison.** Conversely an isomorphism $\Gamma \rightarrow T$ is the image of a unique original cell s by [theorem 6.4](#).
- (3) **Verify contextual universality.** Whiskering it by any L, R is an isomorphism. Precomposition with that isomorphism is a bijection into every atomic output B . Apply [equation \(6.2\)](#) to both its source profiles; the resulting bijection is [equation \(5.2\)](#).
- (4) **Test a specified comparison.** The same argument applies to any specified s whose image is invertible. $\square$

Theorem 6.6 (Universal property). *An interpretation of P in the single-output part of a strict path 2-category extends to $L(P)$ if and only if it sends every member of $S(P)$ to an invertible comparison. The extension is unique once its values on the original signature are fixed.*

Proof.

- (1) **Extend to forests.** Extend first to forests by [lemma 5.2](#).
- (2) **Prove necessity.** Necessity follows from [equation \(6.1\)](#).
- (3) **Define the inverse images.** For sufficiency send s^{-1} to the inverse of the interpreted s ; all defining equations hold.
- (4) **Prove uniqueness.** An inverse, when it exists, is unique, so all values of the extension are forced. $\square$

Remark 6.7. The backward evaluator is a normalization theorem for cells with one atomic output. It makes no assertion that arbitrary many-output cells have a forest normal form. Such an assertion is unnecessary for [theorems 6.4](#) to [6.6](#).

7. CHANGING VERTICAL BOUNDARIES

Fix a virtual equipment $\mathbb{D}$, and let P be its entire globular calculus. The constructions below use only represented units and restrictions already present in $\mathbb{D}$. For the displayed calculations, choose representatives for the finitely many objects and vertical arrows involved and write $f_* : X \rightarrowtail X'$ and $f^* : X' \rightarrowtail X$ for the resulting companion and conjoint. No simultaneous choice of such representatives is part of the construction: [proposition 7.4](#) below quotients all

local presentations by their unique binding-compatible comparisons. This is the familiar binding or base-change calculus; compare [Shu08, Sections 4–5].

Lemma 7.1 (Binding and restriction comparisons). *The units and restrictions of $\mathbb{D}$ give the following data in $L(P)$.*

(1) For $f : X \rightarrow X'$, an adjunction $f_* \dashv f^*$ with

$$j_f : \emptyset_X \Rightarrow f_* f^*, \quad e_f : f^* f_* \Rightarrow \emptyset_{X'},$$

satisfying

$$\begin{array}{ccc} f_* & \xrightarrow{j_f \odot 1} & f_* f^* f_* \\ & \searrow & \downarrow 1 \odot e_f \\ & & f_* \end{array} \quad \begin{array}{ccc} f^* & \xrightarrow{1 \odot j_f} & f^* f_* f^* \\ & \searrow & \downarrow e_f \odot 1 \\ & & f^* \end{array} \quad (7.1)$$

(2) For $H : X' \nrightarrow Y'$ and $f : X \rightarrow X'$, $g : Y \rightarrow Y'$, an intrinsically universal comparison, with unit factors suppressed,

$$s_{f,H,g} : f_* H g^* \Rightarrow H[f, g]. \quad (7.2)$$

(3) For composable $f : X \rightarrow Y$, $u : Y \rightarrow Z$, invertible comparisons

$$k_{f,u} : f_* u_* \Rightarrow (uf)_*, \quad l_{f,u} : u^* f^* \Rightarrow (uf)^*.$$

They are associative and unital, and identify the displayed adjunction for uf with the composite adjunction.

Proof.

- (1) **Construct the binding pairs.** We recall the binding argument, to specify which universal properties are used. The cartesian cell $f_* \Rightarrow U_{X'}$ has sides $(f, 1)$. Factoring the unit cell on f through it gives the other companion binding cell, with sides $(1, f)$. The corresponding two cells for f^* have the reversed sides. A vertical paste of each pair is the unit square on f .
- (2) **Prove the horizontal binding identity.** The horizontal binding identity is proved by postcomposing with the defining cartesian cell: both sides then give that same unit square, and uniqueness of cartesian factorization makes them equal.
- (3) **Construct the restriction comparison.** Paste the cartesian companion and conjoint cells on the two ends of H , and remove the displayed copies of $U_{X'}$ and $U_{Y'}$ by their unit universal properties. Factoring through $H[f, g]$ gives [equation \(7.2\)](#).
- (4) **Prove contextual universality.** To factor a cell having $f_* H g^*$ in its source, insert the other two binding cells on its sides and then use the cartesian factorization defining $H[f, g]$. Substitution back gives the original cell by the horizontal binding identities. In the opposite order, cartesian uniqueness gives the identity factorization. This works inside any L, R and with arbitrary exterior maps; hence the comparison belongs to $S(P)$. This is the binding form of the opcartesian base-change comparison of [CS10, Theorem 7.16].
- (5) **Construct the adjunction unit.** For the first assertion, the same argument with $H = U_{X'}$ makes $f_* f^*$ a representative of $U_{X'}[f, f]$, after removal of a unit factor. Let $t_f : \emptyset_X \Rightarrow U_{X'}[f, f]$ be the factor of the nullary unit at X' with sides (f, f) . Define j_f by composing t_f with the inverse representing comparison.
- (6) **Construct the adjunction counit.** The two cartesian binding cells, followed by unit composition, give $f^* f_* \Rightarrow U_{X'}$; compose with the inverse of $\emptyset_{X'} \Rightarrow U_{X'}$ to obtain e_f .

- (7) **Verify the triangle identities.** The two composites in [equation \(7.1\)](#), on expanding these definitions, are the horizontal binding pastes just proved. Removal and insertion of the representing comparisons cancel. Thus they are the identity cells.
- (8) **Construct composite binding comparisons.** For composable $f : X \rightarrow Y$ and $u : Y \rightarrow Z$, paste their companion binding pairs. Factoring the resulting binding pair through a companion binding pair for uf gives a comparison $k_{f,u} : f_*u_* \Rightarrow (uf)_*$. By [\[CS10, Corollary 7.19\]](#), the pasted companion path has an opcartesian composite comparison to the base-change object of uf . That comparison is characterized by the same pasted binding cells as $k_{f,u}$, so cartesian uniqueness identifies it with the comparison just constructed. Its factorization property holds in every surrounding horizontal context; hence [lemma 5.5](#) places $k_{f,u}$ in $S(P)$, and it is invertible in $L(P)$. Dually, the pasted conjoint binding pairs give $l_{f,u} : u^*f^* \Rightarrow (uf)^*$, which likewise belongs to $S(P)$ and is invertible in $L(P)$.
- (9) **Prove comparison coherence.** Two comparisons between binding pairs which respect the binding cells are equal: postcompose with a cartesian binding cell and use its uniqueness. For three composable vertical arrows, the two composites of the k 's carry the same pasted companion binding pair to the binding pair of the threefold composite, so they agree; the same argument applies to the l 's. Identity comparisons are proved similarly.
- (10) **Identify composite adjunctions.** Finally the constructions of j_f, e_f from the binding cells show, again by uniqueness, that these comparisons identify the displayed adjunction for a composite with the composite adjunction. $\square$

We normalize a local presentation of a vertical identity to the empty path with identity adjunction. The comparison $\emptyset_X \cong U_X$ permits this normalization without changing the atomic graph. For later calculations, compatibility with composition in [lemma 7.1](#) says explicitly

$$j_{uf} = (k_{f,u} \odot l_{f,u})(1_{f_*} \odot j_u \odot 1_{f^*})j_f, \quad (7.3)$$

$$e_{uf} = e_u(1_{u^*} \odot e_f \odot 1_{u_*})(l_{f,u}^{-1} \odot k_{f,u}^{-1}). \quad (7.4)$$

Definition 7.2 (Cloven global merge presentation). Given a system $\mathbf{b}$ of represented units, companions, and conjoints, define an auxiliary presentation $\mathbf{E}_{\mathbf{b}}(\mathbb{D})$. Its objects and vertical arrows are those of $\mathbb{D}$, and its horizontal boundaries are paths in the original horizontal graph. Define

$$\mathbf{E}_{\mathbf{b}}(\mathbb{D})_{f,g}(\Gamma; \Delta) := L(P)(\Gamma; f_*\Delta g^*). \quad (7.5)$$

For a square a , denote its right-hand representative by $\widehat{a}$. Horizontal pasting of a over (f, g) and b over (g, h) is

$$\begin{array}{ccc} \Gamma\Gamma' & \xrightarrow{\widehat{a} \odot \widehat{b}} & f_*\Delta g^*g_*\Delta'h^* \\ & \searrow \widehat{a \odot b} & \downarrow 1 \odot e_g \odot 1 \\ & & f_*\Delta\Delta'h^*. \end{array} \quad (7.6)$$

If b below a has sides (u, v) , define vertical pasting by

$$\begin{array}{ccc} \Gamma & \xrightarrow{\widehat{a}} & f_*\Delta g^* \\ \widehat{ba} \downarrow & & \downarrow 1 \odot \widehat{b} \odot 1 \\ (uf)_*\Xi(vg)^* & \xleftarrow{k_{f,u} \odot 1 \odot l_{g,v}} & f_*u_*\Xi v^*g^* \end{array} \quad (7.7)$$

where Ξ is the lower boundary of b . The vertical identity on Γ is 1_Γ ; the horizontal identity on f is j_f .

Theorem 7.3 (Pasting coherence). *For every cloven presentation $\mathfrak{b}$, the operations of [definition 7.2](#) give $E_{\mathfrak{b}}(\mathbb{D})$ the structure of a strict double category with paths as horizontal arrows, and hence a merge calculus on the original atomic graph.*

Proof.

- (1) **Prove horizontal associativity.** For three horizontally adjacent squares, either association tensors their representatives and contracts the two disjoint strings g^*g_* and h^*h_* . The contractions commute by interchange in $L(P)$.
- (2) **Verify horizontal identities.** Horizontal composition with j_f or j_g is an identity by the two triangle identities [equation \(7.1\)](#).
- (3) **Prove vertical associativity.** For three vertically adjacent squares, either association first substitutes all three representatives and then compares the three successive companions and the three successive conjoints with those of the composite vertical maps. Associativity of k, l from [lemma 7.1](#) makes these results equal.
- (4) **Verify vertical identities.** Their identity comparisons are normalized, proving the vertical identity laws. [Equation \(7.3\)](#) states exactly that the horizontal identity on uf is the vertical paste of those on f and u .
- (5) **Prove interchange.** It remains to check interchange. At an interior vertical side, write its upper and lower segments as g and v . After expanding the four square representatives, either order of evaluation has the same uncontracted string. Pasting horizontally first contracts g^*g_* , then v^*v_* , in that string. Pasting vertically first replaces the string $v^*g^*g_*v_*$ by $(vg)^*(vg)_*$ using l, k , and contracts by e_{vg} . These operations are equal by [equation \(7.4\)](#). Every other part of the two expressions agrees by functoriality of whiskering. This proves interchange. It also proves compatibility with structural subdivisions and hence evaluation of all rectangular schemes. $\square$

Proposition 7.4 (Choice-free global merge realization). *For each vertical arrow f , let $\mathcal{B}(f)$ be the collection of all binding presentations furnished by [lemma 7.1](#), obtained from any represented unit and the corresponding restrictions. If $\mathfrak{f}, \mathfrak{f}' \in \mathcal{B}(f)$, there are unique binding-compatible isomorphisms*

$$\kappa_{\mathfrak{f}, \mathfrak{f}'} : \mathfrak{f}_* \Longrightarrow \mathfrak{f}'_*, \quad \lambda_{\mathfrak{f}, \mathfrak{f}'} : \mathfrak{f}^* \Longrightarrow \mathfrak{f}'^*,$$

and they satisfy the identity and cocycle equations. The cell collections

$$E(\mathbb{D})_{f,g}(\Gamma; \Delta) := \left(\coprod_{\substack{\mathfrak{f} \in \mathcal{B}(f) \\ \mathfrak{g} \in \mathcal{B}(g)}} L(P)(\Gamma; \mathfrak{f}_* \Delta \mathfrak{g}^*) \right) / \sim \quad (7.8)$$

where

$$(\mathfrak{f}, \mathfrak{g}, a) \sim (\mathfrak{f}', \mathfrak{g}', a') \iff a' = (\kappa_{\mathfrak{f}, \mathfrak{f}'} \odot 1_{\Delta} \odot \lambda_{\mathfrak{g}, \mathfrak{g}'}) a, \quad (7.9)$$

form a merge realization on the original boundary signature. Every finite rectangular pasting is evaluated by choosing representatives for its finitely many cells and vertical seams, evaluating by the local cloven pasting diagrams [\(7.6\)](#) and [\(7.7\)](#), and taking the class in [equation \(7.8\)](#). The result is independent of all such local choices. Thus $E(\mathbb{D})$ is defined without a simultaneous choice of units, companions, or conjoints. Any cloven presentation $E_{\mathfrak{b}}(\mathbb{D})$ is canonically isomorphic to $E(\mathbb{D})$, and the canonical isomorphisms between cloven presentations satisfy the cocycle identity.

Proof.

- (1) **Compare represented units.** For two represented units, universal factorization gives a unique isomorphism carrying one nullary universal cell to the other.

- (2) **Compare binding pairs.** For two companion binding pairs, factor the cartesian binding cell of the first through that of the second and conversely. Each resulting composite has the same factorization through a cartesian binding cell as the identity, so cartesian uniqueness makes the two factors inverse. This gives $\kappa_{f,f'}$; the conjoint argument gives $\lambda_{f,f'}$.
- (3) **Prove the cocycle identities.** A composite of two binding-compatible comparisons is binding-compatible, so uniqueness gives

$$\kappa_{f',f''}\kappa_{f,f'} = \kappa_{f,f''}, \quad \lambda_{f',f''}\lambda_{f,f'} = \lambda_{f,f''},$$

together with the identity equations. Hence [equation \(7.9\)](#) is an equivalence relation.

- (4) **Preserve the binding operations.** Binding-compatible isomorphisms preserve j and e : after expanding these cells by their defining binding factorizations, both possible transports have the same composite with the relevant cartesian binding cell, so uniqueness identifies them. They likewise preserve k and l , since the transported and untransported composites respect the same pasted binding pairs.
- (5) **Make pasting independent of choices.** Therefore changing any representative or seam in a finite rectangular pasting transports its cloven value exactly by the relation [equation \(7.9\)](#). The class of the value is independent of the choices. Only finitely many representatives are needed for a finite scheme, so this definition requires no simultaneous cleavage.
- (6) **Descend the double-category laws.** Evaluate the two sides of any associativity, identity, or interchange law using the same finite choices. The calculation in the proof of [theorem 7.3](#) uses only those finitely many choices and makes the two values equal; hence they determine the same quotient class. Thus the quotient is a strict double category and a merge calculus.
- (7) **Recover each cloven presentation.** If a cloven presentation $\mathbf{b}$ is given, send $a \in L(P)(\Gamma; f_*\Delta g^*)$ to its class. Every quotient class has a unique representative after transport to the $\mathbf{b}$ -presentation of its two exterior arrows, so this map is bijective on every cell collection and respects all pasting. It is therefore a canonical isomorphism $E_{\mathbf{b}}(\mathbb{D}) \cong E(\mathbb{D})$. The cocycle identity is the one already proved for the binding-compatible comparisons. $\square$

Theorem 7.5 (Global conservativity). *There is a virtual functor $\mathbb{D} \rightarrow E(\mathbb{D})$, identical on its boundary signature, inducing bijections*

$$\mathbb{D}_{f,g}(\Gamma; H) \cong E(\mathbb{D})_{f,g}(\Gamma; H) \tag{7.10}$$

for every atomic H . It preserves and reflects original cartesian cells.

Proof.

- (1) **Select local representatives.** For the fixed profile in question, choose binding presentations of the two exterior arrows. By [proposition 7.4](#), every class in $E(\mathbb{D})_{f,g}(\Gamma; H)$ has a unique representative in the corresponding summand of [equation \(7.8\)](#).
- (2) **Construct the cell bijection.** The desired bijection is therefore the top edge of

$$\begin{array}{ccc} \mathbb{D}_{f,g}(\Gamma; H) & \xrightarrow{\cong} & E(\mathbb{D})_{f,g}(\Gamma; H) \\ \cong \downarrow & & \downarrow \cong \\ P(\Gamma; H[f, g]) & \xrightarrow{\cong} & L(P)(\Gamma; f_*Hg^*). \end{array}$$

The left edge is cartesian factorization. The bottom edge is the conservativity bijection of [theorem 6.4](#) followed by postcomposition with $s_{f,H,g}^{-1}$. The right edge selects the local representative of the quotient class. The construction is independent of the two local binding presentations by [equation \(7.9\)](#).

- (3) **Preserve identities.** Identities are preserved by the binding identities.
- (4) **Preserve virtual substitution.** For substitution of a compatible family, choose binding presentations for the finitely many vertical seams and apply the displayed construction to every cell. At each internal side the two binding cells are adjacent and contract by e_g , as in [equation \(7.6\)](#); at the exterior sides their vertical composites are compared by k, l , as in [equation \(7.7\)](#). The horizontal binding identity cancels each inserted pair. The remaining cartesian factor is precisely that of the original substituted cell. Passing to quotient classes proves compatibility with all virtual substitution, including empty blocks.
- (5) **Preserve and reflect cartesianness.** Finally, the test for an original unary cartesian cell is a bijection of original single-output cell collections for every top path and exterior boundary. The displayed bijections identify all these tests and their substitution maps in the two calculi, proving both preservation and reflection. $\square$

Definition 7.6 (Equipment boundary transport). A **boundary presentation** of a vertical arrow $f : X \rightarrow X'$ consists of arrows $f_* : X \rightarrow X'$ and $f^* : X' \rightarrow X$ (empty paths are permitted for identities) and cells

$$j_f : \emptyset_X \Longrightarrow f_* f^*, \quad e_f : f^* f_* \Longrightarrow \emptyset_{X'}$$

satisfying [equation \(7.1\)](#). A poly-virtual calculus has **equipment boundary transport** when every vertical arrow admits a boundary presentation and, for every local choice of presentations for f, g , there are bijections

$$\mathbb{P}_{f,g}(\Gamma; \Delta) \cong \mathbb{P}_{\text{glob}}(\Gamma; f_* \Delta g^*).$$

For two local presentations of the same vertical arrow, these representability bijections induce the unique comparison isomorphisms between their companion boundaries and between their conjoint boundaries: with the other exterior map an identity and the lower path empty, the two companion boundaries represent the same cell functor, and the conjoint case is dual. The boundary-bending bijections are required to be invariant under these canonical comparisons, which also carry the corresponding j, e cells to one another. There are also natural identifications between the cell collections obtained by iterated boundary bending and by bending along composite vertical maps. Coherence means that all diagrams of these bijections generated by inserting identity vertical maps, reassociating a composable string of vertical maps, changing local boundary presentations by their canonical comparisons, and the defined tree substitutions commute. This is coherence of functions on cell collections; it does not declare a multiwire merge to be a tree operation. A **poly-virtual equipment** is a poly-virtual calculus having this property. Transported boundaries such as $f_* \Delta g^*$ need not have atomic representatives. In the realizations constructed below the comparison and coherence clauses are consequences of [lemma 7.1](#) and [propositions 7.4](#) and [7.8](#), not additional structure imposed on the construction.

Proposition 7.7 (Mates in the reduct). *For any local choices of binding presentations for f and g , the realization and its reduct have coherent bijections*

$$\begin{array}{ccc} \mathbb{E}(\mathbb{D})_{f,g}(\Gamma; \Delta) & \xrightarrow{\cong} & L(P)(\Gamma; f_* \Delta g^*) \\ & \searrow \cong & \downarrow \cong \text{ mates} \\ & & L(P)(f^* \Gamma g_*; \Delta). \end{array} \quad (7.11)$$

The second bijection and its inverse use only single-wire cuts. Together with [proposition 7.8](#), these bijections give $U_{\text{poly}} \mathbb{E}(\mathbb{D})$ equipment boundary transport.

Proof.

- (1) **Choose the quotient representative.** The first bijection chooses the unique representative of the quotient class in the selected summand of [equation \(7.8\)](#); equivalently it is [equation \(7.5\)](#) in that cloven presentation.
- (2) **Construct the mate.** For the second, cut the leftmost exposed f_* output against e_f , adjoining the input f^* ; cut the rightmost g^* output against e_g , adjoining g_* . In merge notation the resulting map is

$$a \longmapsto (e_f \odot 1_\Delta \odot e_g)(1_{f^*} \odot a \odot 1_{g_*}).$$

- (3) **Construct and verify the inverse.** For its inverse, cut the f^* output of j_f into the leftmost input and the g_* output of j_g into the rightmost input. Each cut satisfies [equation \(3.1\)](#). The two composites reduce to identity by [equation \(7.1\)](#).
- (4) **Prove substitution and composition compatibility.** Their naturality under tree substitution follows by substituting the same cuts into the corresponding trees. Compatibility with vertical composition is [equations \(7.3\)](#) and [\(7.4\)](#). All these operations and equalities are retained by the reduct. $\square$

Proposition 7.8 (Finite coherence for boundary transport). *For a composable finite string $f_1, \dots, f_n$, iterated use of the comparisons k gives a comparison*

$$K_{f_1, \dots, f_n} : f_{1*} \cdots f_{n*} \Longrightarrow (f_n \cdots f_1)_*,$$

and iterated use of l gives

$$L_{f_1, \dots, f_n} : f_n^* \cdots f_1^* \Longrightarrow (f_n \cdots f_1)^*.$$

These comparisons are independent of all bracketings. Every iterated boundary-bending bijection in [definition 7.6](#) reduces to a common normal form built from j, e, K, L , and all the coherence diagrams required there commute. Hence no further boundary-transport coherence equations are imposed in the realization.

Proof.

- (1) **Normalize composite comparisons.** For three arrows, independence of bracketing is the associativity of k and l proved in [lemma 7.1](#); induction gives the assertion for any finite string. The identity cases use the normalized empty-path presentations.
- (2) **Expand boundary bending.** Expand an iterated bending into insertions of j , cuts against e , and the comparison cells joining successive companion or conjoint paths. Reassociate the latter to K and L . There are four possible local changes of evaluation order.
- (3) **Commute disjoint cuts.** Cuts on disjoint wires commute by interchange.
- (4) **Cancel adjacent unit and counit cells.** Adjacent j, e pairs cancel by the triangle identities [equation \(7.1\)](#).
- (5) **Rebracket binding strings.** Rebracketings of companion or conjoint strings agree by the just-proved independence of K, L .
- (6) **Combine successive boundary maps.** Replacing bending successively along f, u by bending along uf is exactly [equations \(7.3\)](#) and [\(7.4\)](#).
- (7) **Identify all evaluation orders.** Thus any two evaluations with the same typed exterior reduce to the same expression. These calculations are spelled out in [section B](#).
- (8) **Descend through changes of presentation.** Finally, changing the local binding presentations transports every term by the unique comparisons of [proposition 7.4](#), so the conclusion descends to the choice-free realization. $\square$

Remark 7.9 (Restrictions, extensions, and laxity). For a path $K : X' \rightharpoonup Y'$, any local boundary presentations for f, g give the restriction boundary f_*Kg^* . In the corresponding cloven presentation its cartesian correspondence is [equation \(7.5\)](#); the other bending in [equation \(7.11\)](#) gives extension transport. For $f : X \rightarrow X'$, the operation on endopaths $R_f(H) = f_*Hf^*$ is lax monoidal, with comparison

$$f_*Hf^*f_*Kf^* \xrightarrow{1 \odot e_f \odot 1} f_*HKf^*$$

and unit j_f . Contractions of disjoint counits prove associativity, and [equation \(7.1\)](#) proves unitality. No inverse for this comparison is supplied or needed. Vertical reversal interchanges the roles of restriction and extension; the resulting transport remains a formal boundary even when it has no bicomodule representative.

8. THE COMPLETION THEOREM AND MAPS OF CALCULI

Theorem 8.1 (Poly-virtual completion). *The construction $E_{\text{poly}}(\mathbb{D}) = U_{\text{poly}}E(\mathbb{D})$ has the properties stated in [theorem 1.2](#). In particular,*

$$\text{FCat}(E_{\text{poly}}(\mathbb{D})) \cong \text{Mon}(\mathbb{D}),$$

where $\text{Mon}(\mathbb{D})$ denotes the category of monoids and monoid homomorphisms in $\mathbb{D}$, and carriers on the left are the original atomic arrows.

Proof.

- (1) **Assemble the calculus and transport.** The tree calculus and transport follow from [theorem 7.3](#) and [propositions 7.4, 7.7](#) and [7.8](#).
- (2) **Identify the monoid data and equations.** Global conservativity is [theorem 7.5](#), and identifies all multiplication, unit, and homomorphism cells. Its compatibility with substitution identifies all their equations and compositions.
- (3) **Pass to the reduct.** Finally apply [theorem 4.3](#). □

Theorem 8.2 (Exact extension criterion). *Let $F : P \rightarrow P'$ be a map of typed globular calculi. It extends, with the same values on atomic arrows and cells, to a map $L(P) \rightarrow L(P')$ if and only if $F(S(P)) \subseteq S(P')$. For a virtual functor between equipments satisfying this criterion, the extension globalizes canonically to the choice-free merge realizations. In cloven presentations it is represented by the unique binding-compatible comparisons of [proposition 7.4](#).*

Proof.

- (1) **Recognize the extension condition.** By [theorem 6.6](#), extension is equivalent to invertibility of every $F(s)$ in $L(P')$. By [theorem 6.5](#), this is precisely membership in $S(P')$.
- (2) **Transport the binding cells.** For globalization, the image of an original unit comparison becomes invertible. Hence the image binding cells define a companion and conjoint presentation for the image vertical map in the target realization.
- (3) **Compare local presentations.** Compare this local presentation with any other one by the unique binding-compatible comparisons of [proposition 7.4](#), and use these at the two ends of the image of [equation \(7.5\)](#).
- (4) **Preserve the two compositions.** Their compatibility with j, e, k, l proves that [equations \(7.6\)](#) and [\(7.7\)](#) are respected.
- (5) **Prove independence and functoriality.** The cocycle identity makes the resulting quotient class independent of the local presentations, and uniqueness proves compatibility with compositions of such functors. □

Proposition 8.3 (Path presentation). *A merge calculus on a horizontal graph is equivalently a strict double category whose horizontal category is the free path category on that graph, with the graph retained as designated atomic arrows.*

Proof.

- (1) **Construct a strict double category.** From a merge calculus, regard a boundary path as a horizontal arrow and use its cell collections as squares. Two-box rectangular evaluations give the two compositions, and scheme substitution gives associativity, units, and interchange.
- (2) **Evaluate rectangular expressions.** Conversely, compose a rectangular expression in a strict double category. The double-category laws identify the generating equivalent schemes, so its value is well defined.
- (3) **Check that the constructions are inverse.** These constructions recover exactly the specified cell collections and pasting operations in either direction. $\square$

This is a presentation equivalence for the strict path realization used here. It does not identify every poly-virtual calculus with a merge calculus, nor assert that forgetting merges is invertible.

9. FORMAL INTERNAL CATEGORIES IN AN ARBITRARY MONOIDAL BASE

Definition 9.1. The **bicomodule poly-virtual equipment** is

$$\mathbf{Q}(\mathcal{V}) = U_{\text{poly}}(\mathbf{E}(\mathbb{C}(\mathcal{V}))^\vee). \quad (9.1)$$

Vertical reversal reverses the vertical category and the upper and lower boundaries of every square, retaining horizontal orientations. A **formal internal category in $\mathcal{V}$** is a formal category in $\mathbf{Q}(\mathcal{V})$ on an atomic bicomodule carrier.

The variance is summarized here; “primitive” refers to cells before localization.

Data	$\mathbb{C}(\mathcal{V})$	Reversed realization
Objects	Comonoids	Comonoids
Vertical arrows	Opposite comonoid maps	Forward comonoid maps
Atomic arrows	Bicomodules	Bicomodules
Unary cells	Opposite equivariant maps	Forward equivariant maps
Primitive arities	Many inputs, one output	One input, many outputs

Theorem 9.2. *For every monoidal category $\mathcal{V}$, [equation \(9.1\)](#) defines a poly-virtual equipment with the displayed variance. Every primitive single-input cell, including a unary changing-base cell, is preserved and reflected. All its cells admit finite expressions in primitive cells, pasting, and inverses of intrinsically universal comparisons.*

Proof.

- (1) **Complete and reverse the calculus.** Apply [proposition 2.3](#), [lemma 2.4](#), and [theorem 8.1](#), then reverse the vertical category and the squares. Under reversal the original companion and conjoint exchange their roles, as do their unit and counit.
- (2) **Identify equipment transport.** Use the second mate description in [equation \(7.11\)](#) to rewrite the reversed square correspondence in the form required by [definition 7.6](#). The triangle and composition identities are preserved by this reversal, and [proposition 7.8](#) gives the required coherent equipment transport.
- (3) **Recover primitive cells.** The cell assertion is the dual of [theorem 7.5](#).

- (4) **Verify the finite presentation.** The finite-expression assertion is [definitions 6.1](#) and [7.2](#); the latter's adjunction cells were themselves constructed as finite expressions in [lemma 7.1](#). The reduct changes none of these collections. $\square$

A multiplication is therefore an element of the constructed collection $\mathbf{Q}(\mathcal{V})((A, A); A)$, and its associativity is an equality between specified finite expressions. The carrier remains the given bicomodule A . This proves the existence and preservation assertions of [theorem 1.1](#); [theorem 4.3](#) supplies its assertion about tree operations. It remains to identify these expressions with actual cotensor multiplication maps whenever the corresponding boundaries are represented.

10. REPRESENTED COTENSORS AND AGUIAR RECOVERY

Definition 10.1 (Stable cotensor cone). For $M : C \nrightarrow D$ and $N : D \nrightarrow E$, a stable cotensor cone is an equalizer

$$\begin{array}{ccc} T & \xleftarrow{i_{M,N}} & M \otimes N \\ & \nwarrow \exists! \bar{\phi} & \uparrow \phi \\ & & Z \end{array} \quad \begin{array}{c} \xrightarrow{\rho_M \otimes 1} \\ \xrightarrow{1 \otimes \lambda_N} \end{array} \quad \begin{array}{c} M \otimes D \otimes N \\ \\ \end{array} \quad (10.1)$$

which remains an equalizer after applying $K \otimes - \otimes L$ for every pair of objects K, L of $\mathcal{V}$. Write $T = M \square_D N$.

This is a property of an existing cone, not a condition on the definition of $\mathbf{Q}(\mathcal{V})$. Aguiar calls a monoidal category regular when all equalizers exist and are preserved in this way [[Agu97](#), Definition 2.1.1].

Lemma 10.2. *A stable cone (10.1) has unique exterior coactions making $i_{M,N}$ equivariant for the exterior coactions of M, N . It is a C – E bicomodule. Its inclusion is a contextually universal cell $(M, N) \implies T$ of $\mathbb{C}(\mathcal{V})$.*

Proof.

- (1) **Construct the induced coactions.** The map $(\lambda_M \otimes 1)i_{M,N}$ satisfies the balance equation in $C \otimes M \otimes N$: this follows by commuting λ_M past ρ_M in [equation \(2.1\)](#). Since $1_C \otimes i_{M,N}$ is the corresponding equalizer, there is a unique λ_T with

$$\begin{array}{ccc} T & \xrightarrow{i_{M,N}} & M \otimes N \\ \exists! \lambda_T \downarrow & & \downarrow \lambda_M \otimes 1 \\ C \otimes T & \xrightarrow{1_C \otimes i_{M,N}} & C \otimes M \otimes N. \end{array}$$

Construct ρ_T similarly.

- (2) **Verify the bicomodule laws.** Postcomposing the two coassociativity maps with $1_C \otimes 1_C \otimes i_{M,N}$ proves their equality, because this map is monic; counitality uses $i_{M,N}$ in the same way. The right coaction and commuting-coaction equations follow by postcomposing with $i_{M,N} \otimes 1_E \otimes 1_E$ and $1_C \otimes i_{M,N} \otimes 1_E$, respectively. All these maps are monic by stability.
- (3) **Factor a contextual balanced map.** Now take a balanced map $b : B \rightarrow \bigotimes L \otimes M \otimes N \otimes \bigotimes R$. Its balance equation at D gives a unique factor

$$\bar{b} : B \longrightarrow \bigotimes L \otimes T \otimes \bigotimes R$$

through $1 \otimes i_{M,N} \otimes 1$, by stability.

- (4) **Reflect the remaining equations.** For every remaining balance or exterior equation for $\bar{b}$, postcompose both sides with the tensor of $i_{M,N}$ and the additional coaction base. The defining equations for λ_T, ρ_T turn the two resulting maps into the corresponding equal maps for b . The postcomposed map is monic, so the equation for $\bar{b}$ holds.
- (5) **Prove the universal bijection.** Conversely such a balanced $\bar{b}$ gives a balanced b by composition. Uniqueness is that of the equalizer factorization. This is exactly [equation \(5.2\)](#) in $\mathbb{C}(\mathcal{V})$. $\square$

Let $\mathbb{B}(\mathcal{V}) = \mathbb{E}(\mathbb{C}(\mathcal{V}))^\vee$ denote the merge calculus before taking its reduct. Under vertical reversal the cone becomes a cell

$$\iota_{M,N} : T \Longrightarrow (M, N),$$

whose underlying map in $\mathcal{V}$ is $i_{M,N}$.

Proposition 10.3 (Conversion by one cut). *For a stable cotensor cone the construction supplies an inverse comparison*

$$\tau_{M,N} : (M, N) \Longrightarrow T$$

in $\mathbb{B}(\mathcal{V})$. In its poly-virtual reduct, the operation

$$\mathbb{Q}(\mathcal{V})(L, T, R; \Delta) \longrightarrow \mathbb{Q}(\mathcal{V})(L, M, N, R; \Delta), \quad a \longmapsto a(1_L, \tau_{M,N}, 1_R) \quad (10.2)$$

is a bijection for all compatible paths. Likewise there is a nullary comparison $\tau_0 : \emptyset_C \Longrightarrow C$ with this property.

Proof.

- (1) **Invert the stable comparison.** By [lemma 10.2](#), the comparison in $\mathbb{C}(\mathcal{V})$ belongs to $S(P)$. Its image is inverted in $L(P)$; after reversal its inverse has the stated profile.
- (2) **Obtain contextual factorization.** Whiskered precomposition with it is a bijection in the merge calculus.
- (3) **Retain the bijection in the reduct.** The map in [equation \(10.2\)](#) is a single cut along T , so the reduct retains this map, its domain and codomain, and their equalities. Consequently it retains the bijection.
- (4) **Include nullary comparisons.** The nullary assertion follows identically from [lemma 2.4](#). $\square$

The inverse comparison identifies two formal boundaries. The cotensor inclusion $i_{M,N}$ remains its original map in $\mathcal{V}$. The next example explains why the input substitution bijection requires the constructed inverse, even after the original output universal property is known.

Example 10.4 (One-sided representation does not suffice; Boolean obstruction). This is the two-truth-value instance of the Boolean obstruction used to distinguish poly-bicategorical cut from merge composition in [[Had19](#), Section 5], augmented here by the empty structural square needed for our calculus. Let horizontal arrows be the two truth values $\perp < \top$ at one object, and give each profile the empty or singleton collection

$$\mathbb{H}(\Gamma; \Delta) = \begin{cases} \{*\}, & \Gamma = \Delta = \emptyset, \\ \{*\}, & \bigwedge \Gamma \leq \bigvee \Delta, \\ \emptyset, & \text{otherwise.} \end{cases}$$

The vertical category is terminal. The empty meet is $\top$ and the empty join is $\perp$; the first case separately supplies the empty structural square required by [definition 3.3](#). It cannot be a premise of a single-wire cut because it has no ports. For the remaining cells, cuts are valid: if

every remaining input is true and every remaining output is false, the first premise forces the cut formula true and the second forces it false, a contradiction. Applying this argument to the wires of a tree defines all tree evaluations. Unary identities exist, and all evaluation equations hold because each cell collection has at most one element. We obtain a poly-virtual calculus, with the adjoined empty structural square affecting none of its nonempty profiles.

The output list $(\top, \perp)$ is contextually represented on the output side by $\top$, since $\top \vee \perp = \top$ in every join context. Nevertheless

$$\mathbb{H}(\top, \perp; \perp) = \{*\}, \quad \mathbb{H}(\top; \perp) = \emptyset.$$

Thus one-sided output representation alone does not give the input substitution bijection used for multiplication. In our construction that bijection is [equation \(10.2\)](#), obtained from a constructed inverse and then retained under the reduct. This distinguishes its proof from an inference based only on a one-sided universal property.

We now compare iterated representatives. Their unique cone comparisons must supply the associator and unitors used by ordinary cotensor multiplication; this will identify the formal category equations.

Lemma 10.5 (Coherence of represented cotensors). *When the stable cotensors of the pairs being composed exist, all parenthesizations of a finite bicomodule path represent the same balanced cone functor. There is a unique comparison between any two of them commuting with their maps to the unparenthesized tensor. These comparisons are isomorphisms and satisfy the pentagon and triangle identities.*

Proof.

- (1) **Identify the triple cone functor.** For a triple, the object $(M \square_D N) \square_E K$, mapped into $M \otimes N \otimes K$, receives a map from B exactly when a map from B into that tensor satisfies both adjacent balance equations. Indeed the first equation uniquely factors it through $i_{M,N} \otimes 1_K$, and the second then becomes the balance equation for the second cotensor. Monicity after tensoring verifies this equivalence, including exterior equivariance, as in [lemma 10.2](#).
- (2) **Construct the associator.** Reversing the order of these two factorizations gives $M \square_D (N \square_E K)$ and the same cone functor. Put $T_L = (M \square_D N) \square_E K$ and $T_R = M \square_D (N \square_E K)$, with cone maps $I_L = (i_{M,N} \otimes 1)i_{M \square_D N, K}$ and $I_R = (1 \otimes i_{N,K})i_{M, N \square_E K}$. The unique mutually inverse factorizations define the associator by

$$\begin{array}{ccc} T_L & \xrightarrow[\sim]{a_{M,N,K}} & T_R \\ & \searrow I_L & \downarrow I_R \\ & & M \otimes N \otimes K. \end{array}$$

The tensor associator is understood on the lower vertex.

- (3) **Extend to finite paths.** Induction establishes the same property for any finite path.
- (4) **Prove the pentagon identity.** Two routes around the pentagon give maps between representatives with the same composite to the unparenthesized fourfold tensor, so uniqueness makes them equal.
- (5) **Identify the stable unit cones.** For units, $\lambda_M : M \rightarrow C \otimes M$ is the equalizer of $\delta_C \otimes 1$ and $1 \otimes \lambda_M$. If $x : B \rightarrow C \otimes M$ equalizes them, apply ε_C to their first tensor factor to obtain

$$x = \lambda_M(\varepsilon_C \otimes 1)x.$$

Counitality proves uniqueness. This formula survives arbitrary tensoring, so the cone is stable. It gives an isomorphism $C \square_C M \cong M$; the right coaction gives $M \square_D D \cong M$.

- (6) **Prove the triangle identities.** Inserting a unit factor and removing it by its counit preserves the universal cone. Thus the two routes in each triangle have the same cone and agree by uniqueness. This also verifies the nullary cases of the induction. $\square$

Theorem 10.6 (Fixed-base recovery). *For any fixed choices of the stable cotensor representatives used to present cotensor composition, the formal-category structures on a C - C bicomodule A in $\mathbf{Q}(\mathcal{V})$ are in bijection with monoid structures on A in the cotensor monoidal category $\mathbf{Bicomod}_C(\mathcal{V})$. Explicitly,*

$$\begin{array}{ccc} (A, A) & \xrightarrow[\sim]{\tau_{A,A}} & A \square_C A \\ & \searrow \mu & \downarrow m \\ & & A \end{array} \quad \begin{array}{ccc} \emptyset_C & \xrightarrow[\sim]{\tau_0} & C \\ & \searrow \eta & \downarrow u \\ & & A. \end{array} \quad (10.3)$$

The inverse formulas are $m = \mu \iota_{A,A}$ and $u = \eta \tau_0^{-1}$, evaluated in the merge presentation and identified with actual maps by unary full faithfulness. In particular this recovers Aguiar internal categories in every regular monoidal category.

Proof.

- (1) **Transport multiplication and unit cells.** By [proposition 10.3](#) and unary full faithfulness from [theorem 9.2](#),

$$\begin{aligned} \mathbf{Q}(\mathcal{V})((A, A); A) &\cong \mathbf{Q}(\mathcal{V})(A \square_C A; A) \cong \mathbf{Bicomod}_C(\mathcal{V})(A \square_C A, A), \\ \mathbf{Q}(\mathcal{V})(\emptyset_C; A) &\cong \mathbf{Q}(\mathcal{V})(C; A) \cong \mathbf{Bicomod}_C(\mathcal{V})(C, A). \end{aligned}$$

These are the formulas [\(10.3\)](#). Their forward operations each use one cut. We verify all equations.

- (2) **Compare the triple cones.** Write $T = A \square_C A$, $T_L = T \square_C A$, $T_R = A \square_C T$, and let

$$I_L = (\iota_{A,A} \odot 1_A) \iota_{T,A} : T_L \Longrightarrow (A, A, A), \quad I_R = (1_A \odot \iota_{A,A}) \iota_{A,T} : T_R \Longrightarrow (A, A, A).$$

The defining cone equality of [lemma 10.5](#) says $I_R a_{A,A,A} = I_L$.

- (3) **Identify the multiplication squares.** Functoriality of cotensor maps, defined by equalizer factorization, says that the squares

$$\begin{array}{ccc} T_L & \xrightarrow{m \square_C 1} & T \\ I_L \downarrow & & \downarrow \iota_{A,A} \\ (A, A, A) & \xrightarrow{\mu \odot 1} & (A, A) \end{array} \quad \begin{array}{ccc} T_R & \xrightarrow{1 \square_C m} & T \\ I_R \downarrow & & \downarrow \iota_{A,A} \\ (A, A, A) & \xrightarrow{1 \odot \mu} & (A, A) \end{array}$$

commute in the merge calculus. To see this without presupposing any monoid laws, compose their upper maps with the cotensor inclusions. The equalities become the defining factorizations for $m \square_C 1$ and $1 \square_C m$, followed by cancellation of $\tau_{A,A} \iota_{A,A} = 1_T$.

- (4) **Preserve and reflect associativity.** Postcomposing with μ therefore transports the two associativity trees to

$$m(m \square_C 1_A), \quad m(1_A \square_C m) a_{A,A,A} : T_L \longrightarrow A. \quad (10.4)$$

Since I_L is invertible in the merge calculus and unary maps embed faithfully, formal associativity is equivalent to equality of the actual maps in [equation \(10.4\)](#).

- (5) **Preserve and reflect the unit laws.** Let $\ell_A : C \square_C A \rightarrow A$ and $r_A : A \square_C C \rightarrow A$ be the unitors from [lemma 10.5](#). Their cone descriptions identify the left unit tree with

$m(u \square_C 1_A) \ell_A^{-1}$ and the right unit tree with $m(1_A \square_C u) r_A^{-1}$. Thus the formal unit laws are precisely

$$m(u \square_C 1_A) = \ell_A, \quad m(1_A \square_C u) = r_A.$$

This proves preservation and reflection of the full monoid structure.

- (6) **Identify internal categories.** Aguiar's internal category is a monoid in this bicomodule monoidal category [Agu97, Theorem 2.2.1 and Definition 2.3.1]; regularity supplies every stable cone used above. $\square$

The recovery proof is local to its displayed stable paths. If all finite powers of A are stably represented, their chosen representatives, including C for the empty power, form a cotensor monoidal category with all bicomodule maps between them. The cone factorizations of lemma 10.5 identify the cotensor of the m th and n th representatives with the $(m+n)$ th and supply its associator and unitors. The same proof therefore gives the correspondence in this generated category. The changing-base and action comparisons below likewise use only the stable paths displayed in their proofs.

Corollary 10.7 (Represented paths globally). *When every bicomodule path is stably represented, composition of its boundaries identifies every completed square with an ordinary equivariant bicomodule map between the represented boundaries. This identification respects mixed bases and all pasting.*

Proof.

- (1) **Represent the square by an actual map.** Replace each boundary by its representative using the invertible cone comparisons. This gives a unary square, and unary full faithfulness identifies it with an actual equivariant map.
- (2) **Prove independence of replacements.** Independence from parenthesization and from the sequence of replacements follows from lemma 10.5.
- (3) **Identify composition over the same boundaries.** For adjacent squares, pasting the representing cones gives the representing cone of their composite; uniqueness therefore makes composition agree. The exterior vertical maps have not been changed during any replacement. $\square$

11. CHANGING-BASE FUNCTORS AND BIMODULE ACTIONS

Lemma 11.1 (Cotensor maps over a base map). *Let $F : M \rightarrow M'$ and $G : N \rightarrow N'$ be equivariant over comonoid maps (f, g) and (g, h) , respectively. For stable cotensors there is a unique equivariant map*

$$F \square_g G : M \square_D N \longrightarrow M' \square_{D'} N'$$

with

$$\begin{array}{ccc} M \square_D N & \xrightarrow{i_{M,N}} & M \otimes N \\ \exists! F \square_g G \downarrow & & \downarrow F \otimes G \\ M' \square_{D'} N' & \xrightarrow{i_{M',N'}} & M' \otimes N'. \end{array} \quad (11.1)$$

These maps respect identities and composition of all three base maps.

Proof.

- (1) **Construct the balance factorization.** Using equivariance, the two balance maps of $(F \otimes G) i_{M,N}$ in equation (11.1) are

$$(F \otimes g \otimes G)(\rho_M \otimes 1) i_{M,N}, \quad (F \otimes g \otimes G)(1 \otimes \lambda_N) i_{M,N}.$$

They agree. The equalizer property gives the unique factorization.

- (2) **Verify exterior equivariance.** Postcompose its proposed exterior equivariance equations with $1 \otimes i_{M',N'}$ or $i_{M',N'} \otimes 1$. The results follow from exterior equivariance of F, G ; stability makes these testing maps monic.
- (3) **Preserve identity maps.** Identity maps satisfy the characterization, so their factor is the identity.
- (4) **Preserve composition.** For successive pairs (F, G) and (F', G') , either factorization has composite with the final inclusion $(F'F \otimes G'G)i_{M,N}$, so uniqueness proves

$$(F' \square_{g'} G')(F \square_g G) = (F'F) \square_{g'g} (G'G). \quad \square$$

Theorem 11.2 (Global category recovery). *In the represented cotensor setting of [theorem 10.6](#), the correspondence is an isomorphism of categories, including homomorphisms over arbitrary comonoid maps. Such a homomorphism is exactly a comonoid map $f : C \rightarrow D$ and an equivariant map $F : A \rightarrow B$ satisfying*

$$Fu_A = u_B f, \quad Fm_A = m_B (F \square_f F). \quad (11.2)$$

Proof.

- (1) **Identify the carrier map.** Unary global full faithfulness from [theorem 9.2](#) identifies the formal carrier square with exactly the equivariant map F .
- (2) **Transport the multiplication equation.** For multiplication, replace the upper boundary (A, A) by $A \square_C A$. Naturality of the representing comparisons is [equation \(11.1\)](#). Consequently the two formal homomorphism trees become the two maps in

$$\begin{array}{ccc} A \square_C A & \xrightarrow{m_A} & A \\ F \square_f F \downarrow & & \downarrow F \\ B \square_D B & \xrightarrow{m_B} & B. \end{array}$$

- (3) **Transport the unit equation.** For units, the primitive map $f : C \rightarrow D$ intertwines the regular coactions and satisfies $\varepsilon_D f = \varepsilon_C$. Thus it is the map between the nullary representatives over f , and the unit tree becomes the left equality in [equation \(11.2\)](#).
- (4) **Identify the functor category.** Full faithfulness reflects both equations. The identity and composite comparisons follow from [lemma 11.1](#), so this is an isomorphism of categories. These are the internal-functor equations in the cotensor formulation [[Agu97](#), Section 4.1]. $\square$

Definition 11.3 (Atomic bimodule action data). Let (X, A) and (Y, B) be formal categories. An A – B bimodule is an atomic arrow $M : X \rightharpoonup Y$ with action cells

$$p : (A, M) \Longrightarrow M, \quad q : (M, B) \Longrightarrow M$$

satisfying the two associativity squares

$$\begin{array}{ccc} (A, A, M) & \xrightarrow{(\mu_A, 1)} & (A, M) \\ (1, p) \downarrow & & \downarrow p \\ (A, M) & \xrightarrow{p} & M \end{array} \quad \begin{array}{ccc} (M, B, B) & \xrightarrow{(q, 1)} & (M, B) \\ (1, \mu_B) \downarrow & & \downarrow q \\ (M, B) & \xrightarrow{q} & M, \end{array} \quad (11.3)$$

the unit triangles

$$\begin{array}{ccccc}
 (A, M) & \xleftarrow{(\eta_A, 1)} & M & \xrightarrow{(1, \eta_B)} & (M, B) \\
 & \searrow p & \parallel & \swarrow q & \\
 & & M & &
 \end{array} \tag{11.4}$$

and the commuting-action square

$$\begin{array}{ccc}
 (A, M, B) & \xrightarrow{(p, 1)} & (M, B) \\
 (1, q) \downarrow & & \downarrow q \\
 (A, M) & \xrightarrow{p} & M.
 \end{array} \tag{11.5}$$

An equivariant bimodule map over homomorphisms $(f, F) : (X, A) \rightarrow (X', A')$ and $(g, G) : (Y, B) \rightarrow (Y', B')$ is a square $\theta \in \mathbb{P}_{f, g}(M; M')$ with

$$\begin{array}{ccc}
 (A, M) & \xrightarrow{p} & M \\
 (F, \theta) \downarrow & & \downarrow \theta \\
 (A', M') & \xrightarrow{p'} & M'
 \end{array}
 \quad
 \begin{array}{ccc}
 (M, B) & \xrightarrow{q} & M \\
 (\theta, G) \downarrow & & \downarrow \theta \\
 (M', B') & \xrightarrow{q'} & M'.
 \end{array} \tag{11.6}$$

Theorem 11.4 (Bimodule comparison). *The reduct preserves and reflects bimodules, their equivariant maps, and composition of these maps. In the represented bicomodule setting they are exactly bicomodules with cotensor action maps*

$$p_0 : A \square_C M \rightarrow M, \quad q_0 : M \square_D B \rightarrow M$$

satisfying the associative, unital, and commuting action equations. Equivariant maps satisfy

$$\theta p_0 = p'_0(F \square_f \theta), \quad \theta q_0 = q'_0(\theta \square_g G).$$

Proof.

- (1) **Retain the formal bimodule equations.** Every expression in equations (11.3) to (11.6) is a tree evaluation, so the first assertion follows exactly as in theorem 4.3. Composition of maps respects these equations by substitution associativity.
- (2) **Represent the two actions.** For the second assertion, apply proposition 10.3 to A, M and M, B and then use unary full faithfulness.
- (3) **Transport action associativity.** The left associativity equation, tested on $(A \square_C A) \square_C M$, becomes

$$p_0(m_A \square_C 1_M) = p_0(1_A \square_C p_0) a_{A, A, M}.$$

The right associativity equation is the same comparison for M, B, B .

- (4) **Transport commuting actions.** The commuting equation, tested on $(A \square_C M) \square_D B$, becomes

$$q_0(p_0 \square_D 1_B) = p_0(1_A \square_C q_0) a_{A, M, B}.$$

- (5) **Transport action units.** The unit equations become the left and right cotensor unitors, by the nullary calculation in theorem 10.6.
- (6) **Transport bimodule maps.** Finally lemma 11.1 transports equation (11.6) to the two displayed equivariant-map equations. Every step is reversible. $\square$

This proves the recovery assertions of [theorem 1.1](#), including changing-base functors and bimodule actions. These actions supply the atomic distributor carriers in the construction that follows. To obtain restrictions along arbitrary formal functors, we also use the finite bicomodule paths already present in the merge realization. The next three sections construct their balanced multicells, their many-sided completion, and the representation of distributor composition.

12. THE VIRTUAL EQUIPMENT OF FORMAL DISTRIBUTORS

The action data of [definition 11.3](#) already describe distributors on actual bicomodules. To put formal categories at the objects of an equipment, we must also construct distributor multicells and restrictions along formal functors. The merge realization supplies the carriers needed for those restrictions. Throughout this section,

$$\mathbb{B}(\mathcal{V}) = \mathbf{E}(\mathbf{C}(\mathcal{V}))^\vee, \quad \mathbf{Q}(\mathcal{V}) = U_{\text{poly}}\mathbb{B}(\mathcal{V}).$$

We use all horizontal paths of $\mathbb{B}(\mathcal{V})$ as carriers while keeping the objects of the new equipment exactly the formal categories already defined on atomic bicomodules.

12.1. The carrier equipment.

Definition 12.1 (Paths as carriers). Let $\mathbb{K}(\mathcal{V})$ have the objects and vertical arrows of $\mathbb{B}(\mathcal{V})$, and let each finite bicomodule path be an individual horizontal arrow of $\mathbb{K}(\mathcal{V})$. Set

$$\mathbb{K}(\mathcal{V})_{f,g}(H_1, \dots, H_n; K) = \mathbb{B}(\mathcal{V})_{f,g}(H_1 \cdots H_n; K). \quad (12.1)$$

Substitution on the left is the corresponding rectangular evaluation on the right. The underlying path of an empty list of carriers is the empty bicomodule path at its specified vertex.

A cell of [equation \(12.1\)](#) has one output **carrier**, even when that carrier contains several bicomodules. This use of the path presentation is part of the construction. For example, composition along a carrier with several bicomodules is a merge operation in the original graph and is an ordinary substitution in $\mathbb{K}(\mathcal{V})$.

Lemma 12.2. *The calculus $\mathbb{K}(\mathcal{V})$ is a virtual equipment. Its horizontal unit at X is the carrier $\emptyset_X$. For any local boundary presentations of $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$, a restriction of a carrier $K : X' \rightarrow Y'$ has carrier*

$$K[f, g] = f_* K g^*. \quad (12.2)$$

Proof.

- (1) **Establish substitution and represented units.** The strict double-category laws of [theorem 7.3](#), retained under vertical reversal, give the virtual substitution laws in [equation \(12.1\)](#). Concatenation represents every list of carriers, including the empty list, so $\emptyset_X$ is the represented unit.
- (2) **Construct the restriction cell.** Use the reversed companion and conjoint data from [theorem 9.2](#). Boundary bending gives

$$\mathbb{B}_{f,g}(H; K) \cong \mathbb{B}_{\text{glob}}(H; f_* K g^*).$$

The cell corresponding to $1_{f_* K g^*}$ is the restriction cell $\chi_{f,K,g} : f_* K g^* \Rightarrow K$, displayed as

$$\begin{array}{ccc} X & \xrightarrow{f_* K g^*} & Y \\ f \downarrow & \Downarrow \chi_{f,K,g} & \downarrow g \\ X' & \xrightarrow{K} & Y' \end{array}$$

- (3) **Verify the full cartesian property.** To verify its full cartesian property, allow exterior arrows $r : Z \rightarrow X$ and $s : W \rightarrow Y$. The comparison maps for composite companions and conjoints identify

$$\begin{aligned}\mathbb{B}_{r,s}(H; f_* K g^*) &\cong \mathbb{B}_{\text{glob}}(H; r_* f_* K g^* s^*) \\ &\cong \mathbb{B}_{\text{glob}}(H; (fr)_* K (gs)^*) \\ &\cong \mathbb{B}_{fr,gs}(H; K).\end{aligned}$$

Under these bijections the map is postcomposition with $\chi_{f,K,g}$. The same calculation applies when H is a concatenation of any list of carriers. This is cartesianness in $\mathbb{K}(\mathcal{V})$. $\square$

12.2. Distributors and balanced multicells.

Definition 12.3 (Formal distributor). Let $\mathbf{A} = (X, A, \mu_A, \eta_A)$ and $\mathbf{B} = (Y, B, \mu_B, \eta_B)$ be formal categories. A formal distributor $M : \mathbf{A} \rightarrow \mathbf{B}$ has a carrier $H_M : X \rightarrow Y$ in $\mathbb{K}(\mathcal{V})$ and cells

$$p_M : (A, H_M) \Rightarrow H_M, \quad q_M : (H_M, B) \Rightarrow H_M$$

satisfying the associativity, unit, and commuting-action diagrams (11.3)–(11.5), with M replaced by H_M and substitution evaluated in $\mathbb{K}(\mathcal{V})$. We call it **bicomodule-carried** when H_M consists of one actual bicomodule.

For a bicomodule-carried distributor these complete diagrams are tree evaluations, so the data and laws are exactly those of [definition 11.3](#). For general path carriers they are merge evaluations in $\mathbb{B}(\mathcal{V})$.

Definition 12.4 (Balanced distributor multicell). Consider distributors $M_i : \mathbf{A}_{i-1} \rightarrow \mathbf{A}_i$ for $1 \leq i \leq n$, $N : \mathbf{B}_0 \rightarrow \mathbf{B}_1$, and formal functors

$$\mathbf{F} = (f, F) : \mathbf{A}_0 \rightarrow \mathbf{B}_0, \quad \mathbf{G} = (g, G) : \mathbf{A}_n \rightarrow \mathbf{B}_1.$$

For $n > 0$, put $\Gamma = (H_{M_1}, \dots, H_{M_n})$. A distributor multicell over $(\mathbf{F}, \mathbf{G})$ is a cell $\alpha \in \mathbb{K}(\mathcal{V})_{f,g}(\Gamma; H_N)$ satisfying the exterior squares, where A_i, B_j denote category carriers:

$$\begin{array}{ccc} (A_0, \Gamma) & \xrightarrow{(p_{M_1,1}, \dots, 1)} & \Gamma \\ (F, \alpha) \downarrow & & \downarrow \alpha \\ (B_0, H_N) & \xrightarrow{p_N} & H_N \end{array} \quad \begin{array}{ccc} (\Gamma, A_n) & \xrightarrow{(1, \dots, 1, q_{M_n})} & \Gamma \\ (\alpha, G) \downarrow & & \downarrow \alpha \\ (H_N, B_1) & \xrightarrow{q_N} & H_N. \end{array} \quad (12.3)$$

For an internal interface, let Γ_i^+ be Γ with the category carrier A_i inserted between H_{M_i} and $H_{M_{i+1}}$. Let r_i apply q_{M_i} and ℓ_i apply $p_{M_{i+1}}$ at that interface. Interior balance is

$$\begin{array}{ccc} \Gamma_i^+ & \xrightarrow{r_i} & \Gamma \\ \ell_i \downarrow & & \downarrow \alpha \\ \Gamma & \xrightarrow{\alpha} & H_N. \end{array} \quad (1 \leq i < n). \quad (12.4)$$

For $n = 0$, the functors have common domain $\mathbf{A} = (X, A)$ and the underlying cell is $\alpha : \emptyset_X \Rightarrow H_N$ over (f, g) . The nullary balance square is

$$\begin{array}{ccc} A & \xrightarrow{(F, \alpha)} & (B_0, H_N) \\ (\alpha, G) \downarrow & & \downarrow p_N \\ (H_N, B_1) & \xrightarrow{q_N} & H_N. \end{array} \quad (12.5)$$

Let $\mathbb{D}_{\text{dist}}(\mathcal{V})$ denote this data, with formal categories as objects, formal functors as vertical arrows, and formal distributors as horizontal arrows. Its multicell collections are denoted $\mathbb{D}_{\text{dist}}(\mathcal{V})_{\mathbf{F}, \mathbf{G}}(M_1, \dots, M_n; N)$.

Proposition 12.5 (Closure under substitution). *The specified data form a virtual double category, and there is an identification*

$$\mathbb{D}_{\text{dist}}(\mathcal{V}) = \text{Mod}_{\text{CS}}(\mathbb{K}(\mathcal{V}))|_{\text{FCat}(\mathbf{Q}(\mathcal{V}))}. \quad (12.6)$$

The restriction in [equation \(12.6\)](#) is full on the displayed objects: all distributors between them are retained. Its vertical category is exactly $\text{FCat}(\mathbf{Q}(\mathcal{V}))$.

Proof.

- (1) **Check balance within a substituted block.** Define substitution by that of $\mathbb{K}(\mathcal{V})$. Consider an inserted category action in a substituted cell. If its position is internal to one substituted block, the balance equation for that block identifies the two results.
- (2) **Check nonempty interfaces.** At the interface between two nonempty blocks, their exterior equations move the action to the corresponding inputs of the outer cell; the outer balance equation then identifies the results.
- (3) **Check empty blocks.** At an empty block, [equation \(12.5\)](#) identifies the actions through its two boundary functors. Repeating this step moves the action past any finite string of empty blocks. It also covers the case in which every block is empty.
- (4) **Check exterior equivariance.** At the two exterior boundaries, the same calculation uses the exterior equations of the outer cell and composition of the boundary functors. Thus the substituted cell satisfies all the defining equations.
- (5) **Verify the virtual-category laws.** Unary identity cells satisfy exterior equivariance, and have no interior balance equations. Associativity and identities now follow from the corresponding identities in $\mathbb{K}(\mathcal{V})$.
- (6) **Identify the module construction.** The definition of a monoid in $\mathbb{K}(\mathcal{V})$ on an original atomic bicomodule is exactly [definition 4.1](#): the comparison with $\mathbf{Q}(\mathcal{V})$ is [theorem 4.3](#). Its homomorphisms are exactly the same pairs (f, F) and equations. Bimodules and their multicells in the module construction are the displayed action, exterior, and balance equations, proving [equation \(12.6\)](#). This is the explicit instance of Leinster’s monoids-and-modules construction [[Lei04](#), Section 5.3], in the formulation of [[CS10](#), Definition 2.8], used here. $\square$

12.3. Regular distributors and restrictions.

Lemma 12.6 (The regular distributor). *For a formal category $\mathbf{A}$, the distributor*

$$U_{\mathbf{A}} = (A, \mu_A, \mu_A)$$

represents the empty distributor path at $\mathbf{A}$ in $\mathbb{D}_{\text{dist}}(\mathcal{V})$. Its universal cell is the original unit η_A .

Proof.

- (1) **Verify the regular distributor.** The monad laws make the displayed data a distributor, and the two unit laws give [equation \(12.5\)](#) for η_A .
- (2) **Insert a unit beside a nonempty context.** We prove the contextual universal property directly. Let $\alpha : (L, R) \Longrightarrow N$ be a distributor multicell whose two paths meet at $\mathbf{A}$. If L is nonempty, insert $U_{\mathbf{A}}$ by the right action on its last distributor. If R is nonempty, use instead the left action on its first distributor. Interior balance for α says that the two formulas agree when both are available.

- (3) **Check the inserted equations.** Associativity of the adjacent action gives balance against the regular action on U_A , and the remaining balance and exterior equations are inherited from α .
- (4) **Treat empty contexts and endpoints.** If both contexts are empty, write the boundary functors as $F = (f, F)$ and $G = (g, G)$. Define the inserted cell by

$$\beta = p_N(F, \alpha) = q_N(\alpha, G).$$

The equality is [equation \(12.5\)](#). The functor multiplication laws and the two action associativities prove the exterior equations for β . Its two possible expressions prove compatibility with both exterior actions. When exactly one context is empty, the corresponding exterior equation supplies the same endpoint calculation.

- (5) **Recover the original multicell.** Substitution of η_A into the inserted factor recovers α by the action unit laws.
- (6) **Prove uniqueness in every context.** Conversely, for a balanced cell β with an inserted U_A , its balance equation at an adjacent distributor, followed by substitution of η_A , forces the insertion formula just given. If both contexts are empty, exterior equivariance and $\mu_A(1, \eta_A) = 1_A$ give $\beta = p_N(F, \beta\eta_A)$; the right formula follows similarly. This proves uniqueness in every context and over every exterior pair of functors. $\square$

Theorem 12.7 (Distributor restriction). *Let $N : A' \rightarrow B'$ be a distributor, and let*

$$F = (f, F) : A \rightarrow A', \quad G = (g, G) : B \rightarrow B'$$

be formal functors. Its restriction $N[F, G]$ exists in $\mathbb{D}_{\text{dist}}(\mathcal{V})$. For any local boundary presentations of f, g , it may be chosen with carrier

$$H = f_* H_N g^*. \tag{12.7}$$

For the ambient cartesian cell $\chi : H \Rightarrow H_N$ over (f, g) , its actions are uniquely specified by

$$\begin{array}{ccc} (A, H) & \xrightarrow{\exists! p_H} & H \\ (F, \chi) \downarrow & & \downarrow \chi \\ (A', H_N) & \xrightarrow{p_N} & H_N \end{array} \quad \begin{array}{ccc} (H, B) & \xrightarrow{\exists! q_H} & H \\ (\chi, G) \downarrow & & \downarrow \chi \\ (H_N, B') & \xrightarrow{q_N} & H_N. \end{array} \tag{12.8}$$

These restrictions are coherent under identities and iterated restriction.

Proof.

- (1) **Construct the restricted actions.** [Lemma 12.2](#) supplies H and χ . Cartesian factorization of the two right sides of [equation \(12.8\)](#) constructs unique globular actions on H .
- (2) **Prove left associativity.** To prove their laws it suffices, by the same cartesian uniqueness, to postcompose with χ . Left associativity is the calculation

$$\begin{aligned} \chi p_H(\mu_A, 1) &= p_N(F \mu_A, \chi) \\ &= p_N(\mu_{A'}(F, F), \chi) \\ &= p_N(F, p_N(F, \chi)) = \chi p_H(1, p_H). \end{aligned}$$

- (3) **Verify the left unit law.** For the left unit, the functor unit equation and the action unit equation give

$$\chi p_H(\eta_A, 1) = p_N(F \eta_A, \chi) = p_N(\eta_{A'}[f], \chi) = \chi.$$

- (4) **Verify the right action laws.** Right associativity and its unit equation use G in the same way.

- (5) **Prove that the actions commute.** For commuting actions,

$$\begin{aligned}\chi q_H(p_H, 1) &= q_N(p_N(F, \chi), G) \\ &= p_N(F, q_N(\chi, G)) = \chi p_H(1, q_H).\end{aligned}$$

Thus H is a distributor, and [equation \(12.8\)](#) says that χ is its equivariant distributor map over (F, G) .

- (6) **Verify distributor cartesianness.** To verify cartesianness in $\mathbb{D}_{\text{dist}}(\mathcal{V})$, take a distributor multicell α with target N whose exterior functors factor through F, G . Its unique underlying factor through χ exists in $\mathbb{K}(\mathcal{V})$. Postcomposing each proposed exterior equation for the factor with χ gives the corresponding equation for α , using [equation \(12.8\)](#). Postcomposing its interior balance equations gives those for α . In the nullary case the same argument gives [equation \(12.5\)](#). Cartesian uniqueness reflects these equalities, so the factor is a distributor multicell. Its uniqueness is already the underlying cartesian uniqueness.
- (7) **Compare successive restrictions.** For further functors $R = (r, R)$ and $S = (s, S)$, the composite ambient cartesian cell identifies the carriers $r_* f_* H_N g^* s^*$ and $(fr)_* H_N (gs)^*$ by the companion and conjoint comparisons. Both induced distributor actions are uniquely characterized by their composites to N , with boundary functors FR and GS . Therefore the comparison is an invertible distributor map.
- (8) **Prove restriction coherence.** The identity comparisons and every association comparison respect the same cartesian cells; uniqueness proves their coherence. $\square$

Corollary 12.8. *For every monoidal $\mathcal{V}$, $\mathbb{D}_{\text{dist}}(\mathcal{V})$ is a virtual equipment with vertical category $\text{FCat}(\mathbb{Q}(\mathcal{V}))$.*

Proof.

- (1) **Assemble the virtual equipment.** Combine [proposition 12.5](#), [lemma 12.6](#), and [theorem 12.7](#). These constructions also verify explicitly the units and restrictions provided abstractly for Mod_{CS} by [\[CS10, Propositions 5.5 and 7.4\]](#). $\square$

The objects selected in [equation \(12.6\)](#) have not been enlarged: their carriers are still actual bicomodules. The horizontal carriers include the paths [equation \(12.7\)](#), with their actions constructed above. Closure under these restrictions is established inside the already defined merge realization.

13. THE POLY-VIRTUAL EQUIPMENT OF FORMAL DISTRIBUTORS

Definition 13.1. The poly-virtual equipment of formal distributors is

$$\text{Dist}_{\text{poly}}(\mathcal{V}) = U_{\text{poly}} \mathbf{E}(\mathbb{D}_{\text{dist}}(\mathcal{V})). \quad (13.1)$$

Its designated atomic horizontal arrows are the individual formal distributors of [definition 12.3](#). A horizontal boundary is a finite path of these distributors, including their intervening formal categories. Each individual distributor also retains its carrier path in the original bicomodule graph.

Thus there are two levels of paths in [equation \(13.1\)](#). The carrier of one distributor is a bicomodule path in $\mathbb{B}(\mathcal{V})$. A boundary in $\text{Dist}_{\text{poly}}(\mathcal{V})$ is a path of distributors. The latter carries the intermediate formal-category actions encoded by [equation \(12.4\)](#).

Theorem 13.2 (Conservative distributor completion). *The construction [equation \(13.1\)](#) has the properties of [theorem 1.3](#). More precisely, it induces natural bijections*

$$\mathbb{D}_{\text{dist}}(\mathcal{V})_{F, G}(\Gamma; N) \cong \text{Dist}_{\text{poly}}(\mathcal{V})_{F, G}(\Gamma; N) \quad (13.2)$$

for every distributor path Γ and individual distributor N . They preserve all distributor substitutions and reflect their equations. Every many-sided cell has a finite expression in distributor multicells, pasting, and inverses of intrinsic contextual comparisons.

Proof.

- (1) **Apply the conservative completion.** Apply [theorems 7.5](#) and [8.1](#) to the virtual equipment of [corollary 12.8](#). These theorems keep its object and vertical categories, which are identified by [proposition 12.5](#), and all its designated horizontal arrows. They give [equation \(13.2\)](#), its substitution compatibility, and the poly-virtual equipment structure.
- (2) **Verify the finite presentation.** The finite-expression assertion is the forest and localization presentation, together with the finite binding construction used to globalize it. Each primitive distributor multicell is itself a cell of $\mathbb{B}(\mathcal{V})$ with the equations of [definition 12.4](#). Thus the whole construction is specified by the existing finite-term calculus and the displayed equations. $\square$

Corollary 13.3 (Preservation of actual bicomodule distributors). *The formal categories, formal functors, and bicomodule-carried distributors form a unital virtual double category with the multicells of [definition 12.4](#). Its inclusion in $\text{Dist}_{\text{poly}}(\mathcal{V})$ preserves and reflects all its single-output cells. In particular, for actual bicomodule carriers M, N , its unary cells are exactly actual equivariant bicomodule maps $\theta : M \rightarrow N$ satisfying [equation \(11.6\)](#).*

Proof.

- (1) **Close the atomic fragment under substitution.** The indicated restriction on horizontal arrows is closed under virtual substitution, which does not form a new carrier. It contains the regular distributor at every object.
- (2) **Apply distributor conservativity.** Its multicells are the same as those of $\mathbb{D}_{\text{dist}}(\mathcal{V})$ on these profiles, so [equation \(13.2\)](#) gives the assertion.
- (3) **Identify unary distributor maps.** For unary cells, [theorem 9.2](#) identifies the underlying square with an actual equivariant bicomodule map, and [equation \(12.3\)](#) become exactly [equation \(11.6\)](#). $\square$

Proposition 13.4 (Functor companions and conjoints). *For a formal functor $F = (f, F) : A \rightarrow B$, any local choice of the restrictions*

$$F_* = U_B[F, 1_B], \quad F^* = U_B[1_B, F]. \quad (13.3)$$

*gives companions and conjoints; their underlying carrier paths are f_*B and Bf^* , respectively. In the completed calculus they have unit and counit*

$$j_F : \emptyset_A \Longrightarrow (F_*, F^*), \quad e_F : (F^*, F_*) \Longrightarrow \emptyset_B,$$

with the triangle identities. There are coherent bijections

$$\text{Dist}_{\text{poly}}(\mathcal{V})_{F,G}(\Gamma; \Delta) \cong \text{Dist}_{\text{poly}}(\mathcal{V})_{\text{glob}}(\Gamma; F_*\Delta G^*). \quad (13.4)$$

They respect the defined tree substitutions and compositions of formal functors.

Proof.

- (1) **Construct the distributor restrictions.** The two restrictions in [equation \(13.3\)](#) exist by [theorem 12.7](#); [equation \(12.7\)](#) computes their carriers.
- (2) **Obtain coherent binding data.** Use these restriction cells and the unit comparison of [lemma 12.6](#) in [lemma 7.1](#), now with input $\mathbb{D}_{\text{dist}}(\mathcal{V})$. That lemma constructs j_F, e_F , proves the triangle identities, and supplies the coherent comparisons for composite functors.

- (3) **Construct the mate bijections.** [Proposition 7.7](#) gives [equation \(13.4\)](#). Its inverse and its second mate description use the single-wire cuts of that proposition, now on distributor wires.
- (4) **Verify composition and substitution compatibility.** The unit and counit composition equations prove compatibility with vertical composition, and tree substitution compatibility follows by substituting the same binding cuts into each tree.
- (5) **Normalize identity companions.** Identity companions may be normalized to empty distributor paths as in [section 7](#). $\square$

Remark 13.5 (The second comparison class). The unit $\eta_A : \emptyset_A \Rightarrow U_A$ is contextually universal in the distributor calculus by [lemma 12.6](#). Its underlying cell $\eta_A : \emptyset_X \Rightarrow A$ in $\mathbb{B}(\mathcal{V})$ need not be invertible. For example, over the unit comonoid in **Set**, take the two-element group as a monoid. Its unit map from the singleton is not an isomorphism; unary conservativity and the represented singleton identify this with noninvertibility of the underlying unit cell in $\mathbb{B}(\mathbf{Set})$.

Forgetting distributor actions defines a virtual functor to $\mathbb{K}(\mathcal{V})$, but this example shows that it does not in general preserve intrinsic contextual comparisons. By [theorem 8.2](#), it therefore need not extend across the second localization with the same values on cells. The second completion uses the distributor universal properties, including its regular units.

The argument applies equally to any merge equipment, presented with paths as horizontal arrows, together with a specified collection of its monoids: take its underlying virtual equipment, form modules, restrict on those objects, and apply the conservative completion. In the present application the merge realization is already part of the construction of $\mathbb{Q}(\mathcal{V})$, and the chosen monoids are exactly its formal categories on actual bicomodules.

14. REPRESENTING DISTRIBUTOR CARRIERS AND COMPOSITION

Lemma 14.1 (Transport of a distributor carrier). *Let $M : A \nrightarrow B$ have carrier H , and let $h : H \Rightarrow T$ be an invertible globular cell in $\mathbb{B}(\mathcal{V})$. There is a unique distributor structure on T making h an equivariant isomorphism from M . Its actions are*

$$p_T = hp_M(1_A, h^{-1}), \quad q_T = hq_M(h^{-1}, 1_B). \quad (14.1)$$

Transport also gives bijections on balanced distributor multicells and preserves all their substitutions.

Proof.

- (1) **Determine the transported actions.** The equations for h to preserve the actions force [equation \(14.1\)](#).
- (2) **Verify the action laws.** Substitution cancels each adjacent $h^{-1}h$ in the associative and commuting-action equations, reducing them to those for M . The unit equations reduce to $h1_H h^{-1} = 1_T$. Thus the actions exist and satisfy all the laws.
- (3) **Transport multicells and their equations.** For a multicell, postcompose with the comparison on its output and substitute the inverse comparisons on its inputs. Each exterior or balance equation becomes the corresponding old equation after these cancellations. The same argument covers [equation \(12.5\)](#).
- (4) **Verify invertibility and substitution compatibility.** Inverse transport uses the inverse comparisons, and substitution compatibility follows from cancellation at every internal carrier. $\square$

Corollary 14.2 (Cotensor recovery for distributor carriers). *A distributor whose carrier path is stably represented by an actual bicomodule is canonically isomorphic to a bicomodule-carried distributor. When its action paths are also stably represented, its structures and equivariant maps are exactly the cotensor bicomodule structures and maps of [theorem 11.4](#). The identifications respect changing bases and balanced distributor multicells.*

Proof.

- (1) **Transport actions and maps to the representative.** Use the inverse cone comparison as h in [lemma 14.1](#). Apply [theorem 11.4](#) to the transported actions, and [lemma 11.1](#) to exterior functors and equivariant maps.
- (2) **Transport multicells.** For multicells, transport every carrier first. The bijection of [lemma 14.1](#) respects their balance and exterior equations.
- (3) **Identify maps from represented sources.** For a stably represented source path, unary full faithfulness further identifies the transported cell with a map from the corresponding cotensor; its equations are exactly equivariance and balancing of that map.
- (4) **Prove independence of parentheses.** Cone uniqueness gives independence of the chosen parentheses in each represented path. $\square$

Definition 14.3 (Represented distributor path). A globular distributor multicell $c : \Gamma \Longrightarrow P$ represents Γ when substitution gives bijections

$$\mathbb{D}_{\text{dist}}(\mathcal{V})(L, P, R; N) \longrightarrow \mathbb{D}_{\text{dist}}(\mathcal{V})(L, \Gamma, R; N) \quad (14.2)$$

for every compatible distributor context L, R and every individual distributor N . These are exactly the intrinsic comparisons of the globular calculus of $\mathbb{D}_{\text{dist}}(\mathcal{V})$. By globular detection, they also satisfy the same property over arbitrary exterior formal functors.

Proposition 14.4 (Represented distributor substitution). *For a representing cell $c : \Gamma \Longrightarrow P$, the completed merge realization makes c invertible. In its poly-virtual reduct,*

$$a \longmapsto a(1_L, c, 1_R) : \text{Dist}_{\text{poly}}(\mathcal{V})(L, P, R; \Delta) \longrightarrow \text{Dist}_{\text{poly}}(\mathcal{V})(L, \Gamma, R; \Delta) \quad (14.3)$$

is a bijection for every compatible distributor path Δ and every exterior pair of formal functors. Its forward operation is a single cut along the distributor P . Conversely an original $c : \Gamma \Longrightarrow P$ is invertible in the completion exactly when it has [equation \(14.2\)](#).

Proof.

- (1) **Apply distributor detection.** Apply [theorems 6.5](#) and [7.5](#) to $\mathbb{D}_{\text{dist}}(\mathcal{V})$.
- (2) **Obtain contextual factorization.** In the merge realization, contextual precomposition with an invertible cell is a bijection.
- (3) **Retain the conversion in the reduct.** The forward map in [equation \(14.3\)](#) is a single-wire tree cut, and the reduct retains its values, all the cell collections, and their equalities. It therefore retains the bijection.
- (4) **Prove the converse.** The converse is the detection theorem for this second comparison class. $\square$

We give a construction of representing distributor cells. The following definition describes a property of an existing coequalizer in the merge realization; the definition of $\text{Dist}_{\text{poly}}(\mathcal{V})$ is independent of its existence.

Definition 14.5 (Contextually preserved action coequalizer). Let $M : \mathbf{A} \rightharpoonup \mathbf{B}$ and $N : \mathbf{B} \rightharpoonup \mathbf{C}$ have carriers H_M, H_N , where the middle formal category has carrier B . An action coequalizer is a fork in the relevant globular hom-category of $\mathbb{B}(\mathcal{V})$,

$$H_M B H_N \xrightarrow[1 \odot p_N]{q_M \odot 1} H_M H_N \xrightarrow{c} T, \quad (14.4)$$

in which c is a coequalizer. It is contextually preserved when, for every compatible pair of bicomodule paths L, R , whiskering the entire fork by L and R again gives a coequalizer in the corresponding globular hom-category. The carrier T is a horizontal path of $\mathbb{B}(\mathcal{V})$.

Theorem 14.6 (Relative tensor construction). *A contextually preserved action coequalizer equation (14.4) gives a unique distributor structure $P : \mathbf{A} \rightharpoonup \mathbf{C}$ on T making $c : (M, N) \Longrightarrow P$ a distributor multicell. This multicell represents (M, N) in $\mathbb{D}_{\text{dist}}(\mathcal{V})$.*

Proof.

- (1) **Construct the exterior actions.** Let A, C be the carriers of the two exterior formal categories. The cell $c(p_M \odot 1_{H_N})$ coequalizes the two action maps with A adjoined on the left. To check this, commute the left A -action on M past its right B -action, and then use the equality defining c . Preservation of the fork by left whiskering with A therefore gives a unique left action p_P with

$$p_P(1_A \odot c) = c(p_M \odot 1_{H_N}), \quad q_P(c \odot 1_C) = c(1_{H_M} \odot q_N), \quad (14.5)$$

where the second action is constructed by the right-hand version of the same argument.

- (2) **Verify action associativity and units.** Every contextual instance of c is an epimorphism, since it is a coequalizer. Precompose the proposed left associativity equation with $1_A \odot 1_A \odot c$. Using equation (14.5), it becomes left associativity for p_M , postcomposed with c . Epimorphy gives the desired equality. The left unit equation is tested with c and reduces to the unit equation for p_M . Right associativity and its unit equation are proved by the symmetric calculation with q_N .
- (3) **Verify commuting actions.** The commuting-action equation is tested with $1_A \odot c \odot 1_C$; it reduces to the equality of the two actions on the disjoint factors H_M, H_N , by interchange. Hence P is a distributor.
- (4) **Identify the distributor comparison.** The two equations in (14.5) are its exterior equivariance equations for c , and the coequalizer equation is its interior balance. They also prove uniqueness of the distributor actions.
- (5) **Factor a balanced multicell.** For contextual universality, let $\alpha : (L, M, N, R) \Longrightarrow Z$ be a globular balanced distributor multicell. Its balance at the middle formal category says exactly that the underlying cell coequalizes the whiskered pair in equation (14.4). The preserved coequalizer therefore gives a unique underlying factor $\bar{\alpha} : (L, P, R) \Longrightarrow Z$.
- (6) **Reflect balance and exterior equations.** To verify any remaining balance or exterior equation for $\bar{\alpha}$, precompose with the corresponding contextual instance of c , also including the extra category carrier in that equation. The result is the corresponding equation for α , using equation (14.5) at the two interfaces adjacent to P . That contextual instance is epic, so the equation holds.
- (7) **Prove the contextual bijection.** Conversely, substitution of c into a distributor multicell is a distributor multicell by proposition 12.5. These two operations are inverse by the coequalizer property. This proves equation (14.2).

- (8) **Allow changing bases.** Globular detection in the virtual equipment $\mathbb{D}_{\text{dist}}(\mathcal{V})$ gives all changing-base instances. No preservation of arbitrary coequalizers or ambient exactness property is used: only the stated contextual preservation of the particular fork (14.4). $\square$

When these representations exist for successive distributor pairs, their iterated composites have the expected coherent associators and unitors. Indeed substitution of intrinsic comparisons is intrinsic by lemma 5.4. Any two representatives of a path therefore have a unique invertible distributor map carrying one representing cell to the other. Every route around a pentagon or triangle carries the same representing cell to the same cell, so uniqueness proves equality. The same factorization property defines maps of represented composites over formal functors and proves their functoriality.

For bicomodule-carried distributors in a represented cotensor setting, the two objects before T in equation (14.4) are represented by $H_M \square B \square H_N$ and $H_M \square H_N$, with the indicated intermediate comonoid bases. Thus the pair is the familiar pair of module actions used to form a relative tensor product. Its contextual coequalizer property is a separate representation property from the stable equalizers used to construct those cotensors. The next section verifies both properties in a nonregular example.

15. EXAMPLES AND THE ROLE OF REGULARITY

The unit and Cartesian bases recover familiar categories. Finite group-like bases in **Ab** recover enriched categories and distributors despite ambient nonregularity. In section 16, instability instead occurs within an action boundary on one actual bicomodule.

Proposition 15.1 (The unit comonoid). *For every monoidal $\mathcal{V}$, formal internal categories over I are exactly ordinary monoids in $\mathcal{V}$.*

Proof.

- (1) **Identify the unit-base coactions.** The counit law forces either I -coaction to be the inverse unitor.
- (2) **Compute stable representatives.** The two maps defining balance over I are therefore equal after the canonical unit identifications. Their equalizer is the identity on $M \otimes N$, which is preserved by every tensor functor. Thus every finite path over I is stably represented by its ordinary tensor product, and the nullary representative is I .
- (3) **Recover ordinary monoids.** Theorems 10.6 and 11.2 identify the resulting structures and fixed-base maps with ordinary monoids and their homomorphisms. $\square$

Proposition 15.2 (Identity category). *Every comonoid C in every monoidal $\mathcal{V}$ has a formal internal category with carrier its regular bicomodule C .*

Proof.

- (1) **Transport the identity monoid.** By lemma 2.4, the comparison $d_C : C \Rightarrow \emptyset_C$ is invertible in the merge realization, with inverse τ_0 . Transport the identity monoid on the empty path along this isomorphism. Explicitly,

$$\eta = \tau_0, \quad \mu = \tau_0(d_C \odot d_C) : (C, C) \Rightarrow C.$$

- (2) **Verify the formal monoid laws.** Substitution cancels the adjacent pairs $d_C \tau_0$, proving associativity and the two unit laws. They remain true in the reduct by theorem 4.3.
- (3) **Identify the diagonal comparison.** Equivalently, the multiplication is the inverse comparison for $\delta_C : C \Rightarrow (C, C)$: the unit cone calculation in lemma 10.5 identifies $C \square_C C$ with C in every monoidal category. $\square$

Proposition 15.3 (Cartesian bases). *For a cartesian monoidal category with finite limits, formal internal categories are ordinary internal categories, with their usual internal functors.*

Proof.

- (1) **Identify bicomodules with internal graphs.** Every object has its unique diagonal comonoid. Counitality forces a left coaction $M \rightarrow C \times M$ to have second component 1_M , so it is equivalent to a map $M \rightarrow C$; the right coaction likewise amounts to a map $M \rightarrow D$. Coassociativity and commuting coactions then follow from the product identities. Thus a bicomodule is an internal graph with its two endpoint maps.
- (2) **Identify cotensors with stable pullbacks.** The balance equalizer of $M \times N \rightrightarrows M \times D \times N$ is exactly the pullback $M \times_D N$. Product with any object preserves this equalizer: a map into the product is a pair of maps and factors exactly when its second component does. The same holds on both sides, so the cone is stable.
- (3) **Recover internal categories and functors.** The maps and equations of [theorems 10.6](#) and [11.2](#) now say precisely that composable arrows compose associatively, identity arrows are units, and functors preserve both operations. $\square$

Proposition 15.4 (A nonregular ambient category). *The monoidal category $(\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z})$ is not regular in Aguiar's sense. For*

$$C = \bigoplus_{i=1}^r \mathbb{Z}e_i, \quad \delta_C(e_i) = e_i \otimes e_i, \quad \varepsilon_C(e_i) = 1,$$

all bicomodule paths over C nevertheless have stable cotensors, with

$$(M \square_C N)_{ik} = \bigoplus_{j=1}^r M_{ij} \otimes N_{jk}. \quad (15.1)$$

Formal categories over C are categories enriched in $\mathbf{Ab}$ on the fixed object set $\{1, \dots, r\}$.

Proof.

- (1) **Exhibit failure of ambient regularity.** The coreflexive pair $r, s : \mathbb{Z} \rightrightarrows \mathbb{Z} \oplus \mathbb{Z}$ given by

$$r(n) = (n, 2n), \quad s(n) = (n, 0)$$

has equalizer zero; projection to the first coordinate is a common retraction. After tensoring with $\mathbb{Z}/2$, the two maps are equal, so their equalizer is $\mathbb{Z}/2$. Hence tensoring does not preserve this equalizer.

- (2) **Decompose bicomodules into matrix entries.** For the specified comonoid write

$$\lambda_M(x) = \sum_i e_i \otimes p_i(x), \quad \rho_M(x) = \sum_j q_j(x) \otimes e_j.$$

Coassociativity gives $p_i p_k = \delta_{ik} p_i$ and $q_j q_l = \delta_{jl} q_j$; counitality gives $\sum_i p_i = 1 = \sum_j q_j$. Commuting coactions gives $p_i q_j = q_j p_i$. Thus $M = \bigoplus_{i,j} M_{ij}$, where $M_{ij} = \text{im}(p_i q_j)$. The converse constructs the coactions from any such decomposition.

- (3) **Compute the matrix cotensor.** Decompose $M \otimes N$ into the summands $M_{ij} \otimes N_{kl}$. The two balance maps insert e_j and e_k , respectively. If $j = k$ they agree. If $j \neq k$, projection to the inserted e_j coordinate takes their difference to the identity on this summand. Therefore the equalizer consists exactly of the summands with $j = k$, giving [equation \(15.1\)](#).
- (4) **Prove stability under tensoring.** This argument remains valid after arbitrary tensoring on either side: finite direct sums are preserved, and the coordinate calculation still gives identity or zero on each summand. It proves preservation of the equalizer fork, not just

preservation of its inclusion. In particular, this stability is a coordinate calculation after tensoring and does not use flatness of any of the groups involved.

- (5) **Recover enriched category composition.** For $A : C \nrightarrow C$, a multiplication consists of maps $A_{ij} \otimes A_{jk} \rightarrow A_{ik}$. A bicomodule map $C \rightarrow A$ consists of maps $\mathbb{Z} \rightarrow A_{ii}$. The cotensor associator is the canonical associator for the direct sums of triple tensor products, by its cone characterization. Thus the monoid equations are precisely bilinear category composition and its identity equations. Apply [theorem 10.6](#) to these stable cones. $\square$

Remark 15.5. The preceding identification includes the variation of the finite object set. A comonoid map $\bigoplus_i \mathbb{Z}e_i \rightarrow \bigoplus_j \mathbb{Z}e_j$ must send each e_i to a group-like element. Writing such an element as $\sum_j a_{ij}e_j$, its equations give $a_j^2 = a_j$, $a_j a_k = 0$ for $j \neq k$, and $\sum_j a_j = 1$. Over $\mathbb{Z}$ exactly one coefficient is 1 and the rest are 0. Thus these comonoid maps are exactly functions of the indexing sets, and [theorem 11.2](#) gives the enriched functors.

In this nonregular example, every cotensor relevant to the indicated finite bases is still represented stably. Its role is to show why a global regularity condition is stronger than the requirements of these category structures. The definition in [definition 9.1](#) also applies to paths without such representatives, using the same formal cell calculus.

Proposition 15.6 (Finite-object enriched distributors). *Apply the distributor construction to the part of $\mathbb{B}(\mathbf{Ab})$ whose objects are finite group-like comonoids $C_S = \bigoplus_{s \in S} \mathbb{Z}e_s$ and whose carrier paths have all vertices of this form. Its objects and vertical arrows are finite-object $\mathbf{Ab}$ -enriched categories and enriched functors. Every distributor is isomorphic to one carried by an actual bicomodule, and these distributors are exactly enriched distributors. Every finite distributor path is represented by its relative tensor product. Consequently, after replacing represented boundaries, the completed many-sided cells are exactly the equivariant maps between these relative tensor products.*

Proof.

- (1) **Restrict to finite group-like bases.** Let $\mathbb{B}_{\text{fin}}$ be the specified part of $\mathbb{B}(\mathbf{Ab})$. The companion and conjoint of a comonoid map $C_S \rightarrow C_T$ are bicomodules with these finite bases, so the restriction paths remain in $\mathbb{B}_{\text{fin}}$. Units remain there as well. Thus its path-carrier calculus is a virtual equipment by the proof of [lemma 12.2](#).
- (2) **Represent carrier paths by matrices.** [Proposition 15.4](#) and its proof, with rectangular indexing sets, identify a bicomodule $C_S \nrightarrow C_T$ with a matrix of abelian groups $(M_{ij})_{i \in S, j \in T}$ and identify cotensor composition with matrix multiplication using direct sums and tensor products. All carrier paths in $\mathbb{B}_{\text{fin}}$ therefore have stable atomic representatives.
- (3) **Identify enriched distributor data.** [Lemma 14.1](#) replaces every distributor carrier by such a representative. Its actions are maps

$$A_{ii'} \otimes M_{i'j} \longrightarrow M_{ij}, \quad M_{ij} \otimes B_{jj'} \longrightarrow M_{ij'},$$

with the associative, unital, and commuting equations of an enriched distributor. The exterior equations on maps give the standard equivariant maps over enriched functors. Balanced multicells have the same identification by [definition 12.4](#) and cotensor representation.

- (4) **Construct the balanced tensor quotient.** For distributors $M : A \nrightarrow B$ and $N : B \nrightarrow C$, define

$$(M \otimes_B N)_{ik} = \frac{\bigoplus_j M_{ij} \otimes N_{jk}}{\langle (mb) \otimes n - m \otimes (bn) \rangle}. \quad (15.2)$$

Here the subgroup is generated by all indicated differences, with $m \in M_{ij}$, $b \in B_{jj'}$, and $n \in N_{j'k}$; the two terms belong to the j' and j summands, respectively. For fixed exterior indices i, k , put

$$X = \bigoplus_{j,j'} M_{ij} \otimes B_{jj'} \otimes N_{j'k}, \quad Y = \bigoplus_j M_{ij} \otimes N_{jk}, \quad Q = (M \otimes_B N)_{ik}.$$

The componentwise coequalizer of the two middle actions has the balanced-map factorization

$$\begin{array}{ccc} X & \xrightarrow[1 \otimes p_N]{q_M \otimes 1} & Y & \xrightarrow{\pi} & Q \\ & & \searrow \alpha & & \downarrow \exists! \bar{\alpha} \\ & & & & Z. \end{array}$$

- (5) **Verify contextual preservation of the quotient.** Tensor product of abelian groups preserves coequalizers in either variable, and finite direct sums preserve them as well. Hence matrix cotensor composition preserves this fork under every compatible matrix context. Replacing path carriers by their representatives transfers this contextual coequalizer property to $\mathbb{B}_{\text{fin}}$.
- (6) **Represent distributor pairs and finite paths.** [Theorem 14.6](#) equips the quotient in [\(15.2\)](#) with its exterior actions and proves that it represents the pair as distributors. The regular distributor represents the empty path by [lemma 12.6](#), so induction gives all finite paths. Coherence and functoriality follow from the unique representing comparisons.
- (7) **Identify cells in the second completion.** In the second completion, replace both boundaries by these representatives and use [equation \(13.2\)](#) to identify the resulting unary square with its ordinary equivariant distributor map. This proves the final assertion, including arbitrary exterior enriched functors. $\square$

In this example a functor $F : A \rightarrow B$ with object function $f : S \rightarrow T$ has the familiar represented companion and conjoint

$$(F_*)_{ij} = B_{f(i),j}, \quad (F^*)_{ji} = B_{j,f(i)}.$$

Indeed these are the matrix cotensors of the carrier paths f_*B and Bf^* in [equation \(13.3\)](#); their actions are exactly the restrictions constructed in [theorem 12.7](#). This comparison is made in the indicated finite-base construction, where every carrier path is represented. The global construction of [equation \(13.1\)](#) also retains distributor carriers that pass through other comonoid bases.

16. A DISTRIBUTOR WITH AN UNREPRESENTED ACTION BOUNDARY

The construction has two steps. First form a category whose own composition paths are stable. Then restrict a stable distributor action along a formal functor. The resulting action remains a well-defined cell, while its input path loses contextual representability.

16.1. The category and the functor.

Definition 16.1 (Triangular data). Let $D = \mathbb{Z}g \oplus \mathbb{Z}x$ with

$$\delta(g) = g \otimes g, \quad \delta(x) = g \otimes x + x \otimes g, \quad \varepsilon(g) = 1, \quad \varepsilon(x) = 0.$$

Set $E = \mathbb{Z}e \oplus D$, where e is group-like. Let $H = \mathbb{Z}a \oplus \mathbb{Z}b$ be the E - E bicomodule

$$\lambda_H(h) = e \otimes h, \quad \rho_H(a) = a \otimes g, \quad \rho_H(b) = b \otimes g + 2a \otimes x. \quad (16.1)$$

Also let $J = \mathbb{Z}t$ with $\lambda_J(t) = e \otimes t$ and $\rho_J(t) = t \otimes g$, and define $\varphi : H \rightarrow J$ by $\varphi(a) = 0$, $\varphi(b) = t$.

Lemma 16.2 (Validity of the triangular data). *The displayed formulas make E a comonoid, H, J bicomodules over E , and φ an equivariant bicomodule map.*

Proof.

- (1) **Verify the comonoid laws.** The comonoid laws for E hold on e, g, x : the two iterated coproducts of x are the sum of the three tensors with one x and two g 's, and the counit laws follow from $\varepsilon(g) = 1, \varepsilon(x) = 0$.
- (2) **Verify the bicomodule laws.** For the right coaction on H , write $\rho_H(h) = h \otimes g + d(h) \otimes x$ with $d(a) = 0, d(b) = 2a$. The equation $d^2 = 0$ proves coassociativity, the first square below. Counitality and commuting coactions follow from the formulas; the laws for J use that e, g are group-like.
- (3) **Verify equivariance of the map.** Right equivariance of φ is the second square, since $\varphi d = 0$; left equivariance follows from the common support e :

$$\begin{array}{ccc}
 H & \xrightarrow{\rho_H} & H \otimes E \\
 \rho_H \downarrow & & \downarrow 1 \otimes \delta_E \\
 H \otimes E & \xrightarrow{\rho_H \otimes 1} & H \otimes E \otimes E
 \end{array}
 \qquad
 \begin{array}{ccc}
 H & \xrightarrow{\varphi} & J \\
 \rho_H \downarrow & & \downarrow \rho_J \\
 H \otimes E & \xrightarrow{\varphi \otimes 1} & J \otimes E.
 \end{array}
 \quad \square$$

Lemma 16.3 (Triangular categories). *For $K = H$ or J , the carrier $A_K = E \oplus K$ defines a formal category, with unit the inclusion of E and multiplication the fold*

$$m_K : E \oplus K \oplus K \longrightarrow E \oplus K, \quad (c, k_1, k_2) \longmapsto (c, k_1 + k_2).$$

Every finite composition path is stably represented; for $n \geq 1$,

$$A_K^{\square_{E^n}} \cong E \oplus K^{\oplus n}.$$

Proof.

- (1) **Compute the stable binary blocks.** Expand $(E \oplus K) \otimes (E \oplus K)$ into its four blocks. The E, E, E, K , and K, E blocks have the stable unit cotensors E, K , and K . The K, K block has stable equalizer zero: projection of the inserted E factor onto its e coordinate takes the two balance maps to zero and the identity, respectively. Thus every equalizing map is zero, also after tensoring.
- (2) **Represent every finite word.** For a longer word, remove regular E factors using their stable unit cones. A block with no K contributes E , one K contributes K , and two or more K 's contribute zero. This proves the asserted representatives and stability.
- (3) **Prove associativity of the fold.** Associativity is the fold square

$$\begin{array}{ccc}
 E \oplus K^{\oplus 3} & \xrightarrow{\nabla_{12}} & E \oplus K^{\oplus 2} \\
 \nabla_{23} \downarrow & & \downarrow m_K \\
 E \oplus K^{\oplus 2} & \xrightarrow{m_K} & E \oplus K.
 \end{array}$$

Here ∇_{12} and ∇_{23} fold the indicated pair of K summands; both routes add all three.

- (4) **Verify units and transport the laws.** The unit equations are the summand inclusions. [Theorem 10.6](#) converts these into formal category laws. $\square$

Corollary 16.4 (The restriction functor). *Write $A = E \oplus H$ and $B = E \oplus J$, with the resulting categories $\mathbf{A}, \mathbf{B}$. The map*

$$F = 1_E \oplus \varphi : A \longrightarrow B$$

defines a formal functor $\mathbf{F} = (1_E, F) : \mathbf{A} \rightarrow \mathbf{B}$.

Proof.

- (1) **Verify equivariance.** By lemma 16.2, F is equivariant.
- (2) **Verify the functor equations.** It preserves the unit and the folds of lemma 16.3 through

$$\begin{array}{ccc}
 E \oplus H^{\oplus 2} & \xrightarrow{m_H} & A \\
 \downarrow 1_E \oplus \varphi^{\oplus 2} & & \downarrow F \\
 E \oplus J^{\oplus 2} & \xrightarrow{m_J} & B
 \end{array}
 \qquad
 \begin{array}{ccc}
 E & \xrightarrow{u_A} & A \\
 & \searrow u_B & \downarrow F \\
 & & B.
 \end{array}$$

Theorem 11.2 gives the formal functor laws.

- (3) **Record the carrier ranks.** The carriers A, B, E have ranks five, four, and three, respectively. $\square$

16.2. An actual bicomodule with a nonzero formal action.

Definition 16.5 (The distributor carrier). Put $N = \mathbb{Z}/4$ and $M = Nu \oplus Nv$, with

$$\lambda_M(u) = e \otimes u, \quad \lambda_M(v) = g \otimes v,$$

and the right coaction of the unit comonoid $\mathbb{Z}$. Let 1 denote the identity formal category at $\mathbb{Z}$.

Proposition 16.6 (The stable action). *The carrier M admits a $\mathbf{B}\text{-}1$ distributor structure. Its action paths (B, M) and (B, B, M) are stably represented by $M \oplus N$ and $M \oplus N \oplus N$, respectively. Its left action is induced by the actual equivariant map*

$$p_{B,0} : M \oplus N \longrightarrow M, \quad ((ru + sv), w) \longmapsto (r + w)u + sv.$$

Proof.

- (1) **Compute the stable action cotensor.** For (B, M) , the regular E block gives M ; $J \otimes Nu$ has mismatched supports and gives zero; and both coactions on $J \otimes Nv$ insert g , giving all of N . These computations persist under arbitrary tensoring.
- (2) **Check action equivariance.** The extra N summand and its image Nu both have left support e , so this is equivariant.
- (3) **Verify associativity.** The triple (B, B, M) is stably represented by $M \oplus N \oplus N$, and the action square becomes

$$\begin{array}{ccc}
 M \oplus N \oplus N & \xrightarrow{1_{M \oplus +}} & M \oplus N \\
 \sigma \downarrow & & \downarrow p_{B,0} \\
 M \oplus N & \xrightarrow{p_{B,0}} & M.
 \end{array}$$

Here $\sigma(m, w_1, w_2) = (p_{B,0}(m, w_2), w_1)$; both routes send $((ru + sv), w_1, w_2)$ to $(r + w_1 + w_2)u + sv$.

- (4) **Verify the action unit.** The unit selects M . Hence this gives an associative, unital formal action p_B .
- (5) **Add the commuting right action.** The right action $q : (M, \mathbb{Z}) \Longrightarrow M$ is the unit comparison and commutes with p_B by its coherence. $\square$

Proposition 16.7 (Restriction on the actual carrier). *The same bicomodule M admits a $\mathbf{A}\text{-}1$ distributor structure, obtained by restriction along F of corollary 16.4. Its left action is defined*

by the outer route

$$\begin{array}{ccc}
 (A, M) & \xrightarrow{(F, 1_M)} & (B, M) \\
 p_A \downarrow & & \sim \downarrow \tau_{B, M} \\
 M & \xleftarrow{p_{B, 0}} & M \oplus N
 \end{array} \tag{16.2}$$

where $\tau_{B, M}$ is the stable comparison already constructed. Equivalently, $p_A = p_B(F, 1_M)$, a tree substitution in $\mathbf{Q}(\mathbf{Ab})$. Its right action is the unit action q of [proposition 16.6](#).

Proof.

- (1) **Identify the restricted carrier.** Since the base map of F is 1_E , restriction retains the actual carrier M .
- (2) **Verify restricted associativity.** Associativity follows by pasting

$$\begin{array}{ccccc}
 (A, A, M) & \xrightarrow{(\mu_A, 1)} & (A, M) & \xrightarrow{p_A} & M \\
 (F, F, 1) \downarrow & & \downarrow (F, 1) & & \parallel \\
 (B, B, M) & \xrightarrow{(\mu_B, 1)} & (B, M) & \xrightarrow{p_B} & M \\
 (1, p_B) \downarrow & & \downarrow p_B & & \parallel \\
 (B, M) & \xrightarrow{p_B} & M & = & M.
 \end{array}$$

The two routes around the outside are the two A -action composites, after using the definition of p_A on the left route.

- (3) **Verify the restricted unit law.** Unitality is the left triangle in

$$\begin{array}{ccccc}
 M & \xrightarrow{(\eta_A, 1)} & (A, M) & \xrightarrow{(F, 1)} & (B, M) \\
 & \searrow & \downarrow p_A & \swarrow p_B & \\
 & & M & &
 \end{array}$$

whose outer triangle is the functor unit law followed by the B -action unit law.

- (4) **Verify commuting actions.** The commuting-action equation follows by substituting F into that for p_B, q . Thus (M, p_A, q) is a formal distributor $A \nrightarrow 1$ on one actual bicomodule. The construction of p_A uses the inverse comparison for the stable path (B, M) . $\square$

16.3. The unstable cone and the contextual obstruction.

Lemma 16.8 (The unstable action cotensor). *The ordinary equalizer $T = A \square_E M$ exists and carries an equivariant $E\text{-}\mathbb{Z}$ bicomodule cone. After tensoring with $R = \mathbb{Z}/2$, the comparison from this cone to the new equalizer is neither monic nor epic. In particular, the action cotensor is not stable.*

Proof.

- (1) **Compute the ordinary cotensor.** The regular block of T is M . The $H \otimes Nu$ block is zero by its supports, while an element

$$a \otimes \alpha v + b \otimes \beta v \in H \otimes Nv$$

is balanced exactly when $2\beta = 0$ in N . Indeed its balance obstruction is $2a \otimes x \otimes \beta v$. Consequently

$$T \cong M \oplus Na \oplus N[2]b \cong (\mathbb{Z}/4)^3 \oplus \mathbb{Z}/2. \quad (16.3)$$

- (2) **Describe its equivariant cone.** Writing z for the last, order-two generator, its cone $i : T \hookrightarrow A \otimes M$ is

$$u \mapsto e \otimes u, \quad v \mapsto g \otimes v, \quad a \mapsto a \otimes v, \quad z \mapsto 2b \otimes v. \quad (16.4)$$

Give u, a, z left support e , and v left support g ; this makes T an E - $\mathbb{Z}$ bicomodule and i an equivariant cone.

- (3) **Exhibit a kernel after tensoring.** After tensoring with $R = \mathbb{Z}/2$, the nonzero class $z \otimes \bar{1}$ maps to zero.
- (4) **Exhibit a new balanced element.** Meanwhile

$$b \otimes v \otimes \bar{1} \quad (16.5)$$

becomes balanced: its obstruction is a multiple of 2. This element cannot lie in the image of $i \otimes 1_R$, whose b coordinate is zero.

- (5) **Compute the canonical comparison.** Thus the tensored cone fails both injectivity and surjectivity onto the new equalizer T_R^{bal} . The comparison is

$$\begin{array}{ccc} T \otimes R & \xrightarrow{i \otimes 1_R} & A \otimes M \otimes R \\ \kappa \downarrow & & \parallel \\ T_R^{\text{bal}} & \xrightarrow{i_R} & A \otimes M \otimes R. \end{array}$$

Both objects on the left have underlying group $(\mathbb{Z}/2)^4$. In the source basis (u, v, a, z) and the target basis given by $e \otimes u, g \otimes v, a \otimes v, b \otimes v$ after reduction, κ has matrix $\text{diag}(1, 1, 1, 0)$. The failure is in this canonical cone, whose image has dimension three. $\square$

Theorem 16.9 (An unrepresented action boundary). *The action boundary (A, M) has no atomic contextual representative. Its formal action operates nontrivially on the new balanced element (16.5).*

Proof.

- (1) **Represent the empty-context functor.** In the original globular calculus P of $\mathbb{C}(\mathbf{Ab})$, the cone i is a cell $(A, M) \Rightarrow T$. It represents the empty-context balanced-map functor: factor a balanced map through its ordinary equalizer, and test equivariance of the factor after $1_E \otimes i$. This map is monic because E is free. The right coaction is the unit coaction.
- (2) **Construct the contextual obstruction.** Let $W = \mathbb{Z}/2$ have left E -coaction $w \mapsto e \otimes w$ and right unit coaction. With R regarded as a $\mathbb{Z}$ - $\mathbb{Z}$ bicomodule, the map

$$\beta : W \longrightarrow A \otimes M \otimes R, \quad \bar{1} \mapsto b \otimes v \otimes \bar{1}$$

satisfies both interior balances and exterior equivariance. It is a cell in $P(A, M, R; W)$ for which the factorization triangle fails:

$$\begin{array}{ccc} W & \xrightarrow{\beta} & A \otimes M \otimes R \\ \downarrow \#h & \nearrow i \otimes 1_R & \\ T \otimes R & & \end{array}$$

The dashed arrow denotes the nonexistent factor: the b coordinate of $\beta(\bar{1})$ is nonzero, while that of every element in the image of $i \otimes 1_R$ vanishes. Hence i fails the contextual test of [definition 5.3](#).

- (3) **Exclude every alternative atomic representative.** Every other contextual representative would first represent the same empty-context functor, so its cone would be i composed with a bicomodule isomorphism:

$$\begin{array}{ccc} S & \xrightarrow[\sim]{\theta} & T \\ & \searrow j & \downarrow i \\ & & A \otimes M. \end{array}$$

Here j is the proposed representing cone. Tensoring θ with R cannot change this failure of factorization. No alternative atom can therefore represent the path. By [theorem 6.5](#), (A, M) is not isomorphic to any atom in the globular localization, or in its reversed merge realization.

- (4) **Evaluate the formal action.** In the forward realization, β is the primitive cell $W \Rightarrow (A, M, R)$. Applying F to its first output replaces b by t . The stable comparison for (B, M) followed by p_B then sends $t \otimes v \otimes \bar{1}$ to $u \otimes \bar{1}$. Thus, in the merge realization, the entire evaluation is

$$\begin{array}{ccccc} W & \xrightarrow{\beta} & (A, M, R) & \xrightarrow{(F, 1, 1)} & (B, M, R) \\ \downarrow \gamma & & \downarrow p_A \odot 1_R & & \downarrow \sim \tau_{B, M} \odot 1_R \\ (M, R) & \xlongequal{\quad} & (M, R) & \xleftarrow[p_{B, 0} \odot 1_R]{} & (M \oplus N, R) \end{array}$$

- (5) **Detect the nonzero result.** Here (M, R) is stably represented by $M \otimes R$. Under this comparison, unary conservativity identifies γ with the actual map $\bar{1} \mapsto u \otimes \bar{1} \neq 0$. The reduct retains the resulting cells and equality. $\square$

Corollary 16.10 (The action on the ordinary cotensor). *The map $p_T : T \rightarrow M$ induced by the formal action is nonzero on the order-two summand of T : it sends its generator z to $2u$.*

Proof.

- (1) **Identify the ordinary cotensor action.** The primitive cone gives a triangle in the merge realization

$$\begin{array}{ccc} T & \xrightarrow{\iota} & (A, M) \\ & \searrow p_T & \downarrow p_A \\ & & M, \end{array}$$

where unary conservativity identifies p_T with the actual map

$$p_T : T \longrightarrow M, \quad (ru + sv, \alpha, \beta) \longmapsto (r + \beta)u + sv, \quad \beta \in N[2],$$

so $p_T(z) = 2u \neq 0$.

- (2) **Exclude contextual evaluation through that cotensor.** Nevertheless, the contextual action above cannot be evaluated by first factoring through $T \otimes R$: that factorization does not exist by [theorem 16.9](#). $\square$

Corollary 16.11 (Restriction does not preserve action stability). *Stability of action cotensors need not survive restriction along a formal functor, even when its base map is the identity and*

both formal categories have stable finite composition paths. This failure occurs for a distributor on one actual bicomodule, whose restricted action boundary has no contextual atomic representative.

Proof.

- (1) **Choose the functor and stable category structures.** Use the functor $F : A \rightarrow B$ of [corollary 16.4](#). [Lemma 16.3](#) gives stable composition paths for both categories.
- (2) **Restrict the stable action.** The B -action of [proposition 16.6](#) has a stable action cotensor; its restriction is the A -action of [proposition 16.7](#).
- (3) **Identify the failure of representation.** [Lemma 16.8](#) and [theorem 16.9](#) give the asserted instability and nonrepresentability. $\square$

Remark 16.12 (A prime-indexed family). Replacing $2, 4$ by p, p^2 gives the same construction for every prime p : use $\rho_H(b) = b \otimes g + pa \otimes x$, $N = \mathbb{Z}/p^2$, and $R = \mathbb{Z}/p$. The problematic summand is $N[p]b$, its generator has cone image $pb \otimes v$, and the reduced comparison again has matrix $\text{diag}(1, 1, 1, 0)$. The proofs above apply with these substitutions.

17. FURTHER APPLICATIONS: GENERALIZED MULTICATEGORIES AND POLYCATAGORIES

The bicomodule application required reversal because its primitive maps have the opposite arity to multiplication. In an original virtual double category the monoid cells already have the required arity. Conservativity therefore gives a direct comparison with generalized multicategories. The distributor application likewise keeps the forward variance of its module multicells when applying the completion for the second time. This section records that comparison and the separate example supplied by Garner's two-sided Kleisli construction.

Definition 17.1 (Raw global forest calculus). For any virtual double category $\mathbb{K}$, define $\text{Path}(\mathbb{K})$ using paths as horizontal arrows. A square with $m > 0$ outputs consists of a subdivision of its upper path into m consecutive, possibly empty blocks, a vertical arrow at each cut vertex, and m original cells whose lateral arrows are those successive cut arrows. A square with no outputs is a structural empty square: its upper path is empty and its two exterior maps are the same vertical arrow. Composition substitutes forests vertically and concatenates them horizontally.

Lemma 17.2. *The raw forest calculus is a strict double category with paths as horizontal arrows, and*

$$\text{Path}(\mathbb{K})_{f,g}(\Gamma; H) = \mathbb{K}_{f,g}(\Gamma; H).$$

It therefore preserves and reflects formal categories and their homomorphisms on atomic carriers.

Proof.

- (1) **Verify substitution and interchange.** Substitution composes the arrows at cut vertices as well as the root cells. Associativity is exactly virtual substitution associativity; substitution on disjoint blocks commutes, proving interchange.
- (2) **Handle nullary and identity squares.** Nullary blocks use the degenerate virtual pastings, and the structural empty square on f is its horizontal identity.
- (3) **Recover the single-output fragment.** For one output there is one block and its two cut arrows are the specified f, g , so the displayed equality follows.
- (4) **Retain monoids and homomorphisms.** All monoid and homomorphism operations lie in this fragment, which proves the last statement. $\square$

Theorem 17.3 (Kleisli recovery). *Let $\mathbb{K} = H\text{-Kl}(\mathbb{X}, T)$ be the horizontal Kleisli virtual double category of a monad on a virtual double category. There is an isomorphism*

$$\text{FCat}(U_{\text{poly}}\text{Path}(\mathbb{K})) \cong T\text{-Mon}(\mathbb{X}).$$

When $\mathbb{K}$ is a virtual equipment, its refined realization $\mathbf{E}_{\text{poly}}(\mathbb{K})$ gives the same comparison and also preserves its represented composites and boundary transport.

Proof.

- (1) **Identify the Kleisli monoid data.** A horizontal Kleisli arrow from X to Y is the original arrow $X \rightarrow TY$ with the Kleisli typing. By the definition of a T -monoid [CS10, Definition 4.3], its multiplication and unit are the monoid cells in $\mathbb{K}$.
- (2) **Pass to the path calculus and reduct.** Apply lemma 17.2 and theorem 4.3.
- (3) **Apply equipment completion when available.** If $\mathbb{K}$ is an equipment, apply theorem 8.1 instead; its inverted comparisons are precisely the intrinsic ones by theorem 6.5. $\square$

Proposition 17.4 (Normalization is tracked). *For a monad T on a virtual equipment $\mathbb{X}$, the preceding comparison restricts to normalized T -monoids. More explicitly, the original unit square $U_X \Rightarrow A$ with sides $(1_X, \eta_X)$ is cartesian in $\mathbb{X}$ if and only if its image is cartesian in $\mathbf{E}(\mathbb{X})$.*

Proof.

- (1) **Identify the normalization test.** Normalization is the cartesianness of this unit square [CS10, Definition 8.2]. Its top is atomic and its bottom is atomic in the original equipment $\mathbb{X}$.
- (2) **Preserve and reflect cartesianness.** Theorem 7.5 preserves and reflects all the single-output factorization tests defining cartesianness.
- (3) **Retain the original unit square.** The object correspondence in theorem 17.3 retains the original unit square as data and therefore retains this property. $\square$

Here cartesianness is tested in $\mathbb{X}$, as in the definition of normalization. No extension of T across a localization is used to make this test. This matters when T itself does not satisfy the extension criterion of theorem 8.2.

Proposition 17.5 (The symmetric-polycategory benchmark). *Garner’s description of symmetric polycategories as monads on discrete objects in a two-sided Kleisli bicategory is a further instance of formal-category data in a poly-virtual calculus.*

Proof.

- (1) **Identify the Kleisli bicategory and its monads.** Garner constructs the two-sided Kleisli bicategory using a pseudo-distributive law whose suitable matchings have associated graphs that are connected, acyclic, and have no multiple edges [Gar08, Proposition 9 and Definition 14], and identifies its monads on discrete objects with symmetric polycategories [Gar08, Theorem 42].
- (2) **Construct its coherent path presentation.** For any bicategory, fix a bracketing of each finite path of 1-arrows and take cells between the corresponding bracketed composites. Rectangular evaluation is composition followed by the canonical associator and unitor comparisons; bicategorical coherence makes the result independent of the evaluation order [Lei04, Section 1.5].
- (3) **Identify monads in the reduct.** A monad on one original 1-arrow is exactly equations (1.1) and (4.1) in this path presentation. Apply theorem 4.3.
- (4) **Retain the given symmetric structure.** The symmetric permutations here are part of Garner’s Kleisli data; this argument does not insert exchanges into our planar wire calculus. $\square$

18. UNIVERSAL CHARACTERIZATION AND COMPARISON WITH OTHER ENVELOPES

Theorem 18.1 (Characterization of the construction). *The globular merge completion is characterized by the following presentation: freely form forests of the entire typed single-output calculus and make every member of $S(P)$, and hence each of its horizontal contextual instances, reversible. It adds no original single-output cells and identifies no such cells. Its poly-virtual reduct retains all cells and equalities, the tree operations, and coherent boundary transport; it retains the entire formal-category and bimodule-action theory. Applied to the input $\mathbb{D}_{\text{dist}}(\mathcal{V})$, the same characterization gives $\text{Dist}_{\text{poly}}(\mathcal{V})$ and preserves all primitive distributor multicells, including their changing-base instances.*

Proof.

- (1) **Identify the presentation and reversible comparisons.** The presentation and its uniqueness are [theorem 6.6](#). The exact recognition of reversible atomic comparisons is [theorem 6.5](#).
- (2) **Apply both conservativity theorems.** The two conservativity assertions are [theorems 6.4](#) and [7.5](#).
- (3) **Pass to the reduct.** For the reduct use [theorems 4.3](#) and [11.4](#) and [proposition 7.7](#).
- (4) **Apply the distributor instance.** For the distributor instance apply [theorem 13.2](#). $\square$

The path presentation in [proposition 8.3](#) is the merge form of the two-sided calculus; compare the distinction between virtual and implicit boundary operations in [\[FS26, Remark 1.1\]](#). An independently defined module or presheaf envelope requires the following comparison.

Proposition 18.2 (A concrete comparison test). *Let $J : P \rightarrow \mathcal{M}$ be a globular interpretation in a path presentation of another envelope. If J sends every $s \in S(P)$ to an invertible comparison, there is a unique induced interpretation $\bar{J} : L(P) \rightarrow \mathcal{M}$. It is an equivalence on a hom-category precisely when it is fully faithful there and every object of that hom-category is isomorphic to an interpreted path.*

Proof.

- (1) **Obtain the extension.** The extension statement is [theorem 6.6](#).
- (2) **Construct a quasi-inverse.** A fully faithful and essentially surjective functor is an equivalence: choose, for each target object, an isomorphism from an object in the image; use full faithfulness to pull back the conjugated morphisms. The resulting functor is a quasi-inverse by those same isomorphisms.
- (3) **Prove necessity and apply the criterion.** Conversely an equivalence is fully faithful and essentially surjective. Applied to $\bar{J}$ this gives the stated test. $\square$

An identification with a module or Yoneda envelope therefore requires an interpretation that satisfies this test. The notation Mod_{CS} used here refers throughout to the monoids-and-bimodules construction. It first supplies the balanced bicomodule calculus from $\mathcal{V}^{\text{op}}$, and then supplies distributor multicells from the path-carrier equipment $\mathbb{K}(\mathcal{V})$. These two uses are explicit and do not identify the result with an independently defined presheaf envelope. In particular, [remark 13.5](#) explains why forgetting distributor actions need not extend across the second localization.

The next concrete comparisons suggested by these results are to construct such an interpretation for a chosen presheaf or module envelope, to study the representations of distributor paths beyond the coequalizer construction of [theorem 14.6](#), and to replace the planar tree shapes by other specified arity systems while retaining an analogue of backward evaluation. An intrinsic

distributor construction starting only from an abstract tree calculus is another question: the construction proved here uses the merge realization already provided for $\mathbf{Q}(\mathcal{V})$.

19. CONCLUSION

Finite bicomodule paths specify category and action boundaries in every monoidal category. Stable cotensors represent those boundaries and recover the usual operations, equations, and changing-base functors. The two conservative completions retain the original balanced bicomodule and distributor cells while making intrinsic contextual representations reversible. Cartesian factorization supplies distributor restrictions; contextually preserved action coequalizers represent their composition.

The examples separate two failures. Finite group-like bases in $\mathbf{Ab}$ have stable cotensors and recover the enriched equipment even though $\mathbf{Ab}$ is not regular. The triangular example has stable category composition but an action boundary with no contextual atomic representative. Its nonzero action is constructed by restriction along a formal functor on the same actual bicomodule. Thus the formal calculus also supports operations beyond the stably represented sector.

APPENDIX A. PASTING CONVENTIONS AND ADMISSIBLE ELEMENTARY SHAPES

The definition using rectangular schemes is finite and determines every admissible operation. The elementary globular cases can also be listed as follows; each formula requires matching vertex labels.

- (1) Input substitution: $\alpha : \Gamma \Rightarrow M, \beta : (L, M, R) \Rightarrow \Delta$ give $(L, \Gamma, R) \Rightarrow \Delta$.
- (2) Output substitution: $\alpha : \Gamma \Rightarrow (L, M, R), \beta : M \Rightarrow \Delta$ give $\Gamma \Rightarrow (L, \Delta, R)$.
- (3) A right corner cut is [equation \(3.2\)](#).
- (4) A left corner cut has premises $\Gamma \Rightarrow (M, \Delta)$ and $(\Lambda, M) \Rightarrow \Sigma$ and conclusion $(\Lambda, \Gamma) \Rightarrow (\Sigma, \Delta)$.

These cases, with their overlaps, are exactly [equation \(3.1\)](#). Arbitrary vertical labels are governed by the rectangular matching rule, not by suppressing the labels in this list. For example, to substitute n cells above a virtual cell one specifies all $n + 1$ interface vertical arrows together; adjacent members must use the very same map at their common vertex.

The equations defining evaluation have an equally finite structural description: deleting structural strips does not change evaluation; evaluating a subtree and replacing it by its value does not change evaluation; and equivalent rectangular subdivisions have the same value. Associativity of nested substitutions is the second rule. For disjoint subtrees it applies twice in either order. All higher instances follow by induction on the number of rectangles. In particular, no equation asserts equality of differently typed pastings or supplies a cut where an exterior directed boundary fails to exist.

APPENDIX B. EXPLICIT ADJUNCTION CALCULATIONS

This appendix records the algebra needed in the global construction. For composable f, u, w , the companion comparisons satisfy

$$k_{uf,w}(k_{f,u} \odot 1_{w*}) = k_{f,wu}(1_{f*} \odot k_{u,w}). \quad (\text{B.1})$$

The conjoint equation is its reversed-order counterpart. Both sides of [equation \(B.1\)](#) carry the pasted binding pair to the binding pair of wuf ; the uniqueness argument in [lemma 7.1](#) therefore proves the equation. Applied to four factors, this same characterization proves every association comparison equal, without an additional coherence choice.

For a globular $a : \Gamma \Rightarrow f_* \Delta g^*$ write

$$\mathbf{m}(a) = (e_f \odot 1_\Delta \odot e_g)(1_{f^*} \odot a \odot 1_{g_*}).$$

For $b : f^* \Gamma g_* \Rightarrow \Delta$, its inverse mate is

$$\mathbf{n}(b) = (1_{f_*} \odot b \odot 1_{g^*})(j_f \odot 1_\Gamma \odot j_g).$$

Expanding $\mathbf{m}(b)$ makes the f terms contract by $(e_f \odot 1_{f^*})(1_{f^*} \odot j_f) = 1_{f^*}$ and the g terms by $(1_{g_*} \odot e_g)(j_g \odot 1_{g_*}) = 1_{g_*}$. The result is b . Expanding $\mathbf{n}(a)$ uses the other two triangle identities and gives a . These expansions also prove naturality: a whiskered cell can be moved past a contraction on a disjoint factor by interchange.

For the interchange calculation in [theorem 7.3](#), the only nontrivial local equality is

$$e_{vg}(l_{g,v} \odot k_{g,v}) = e_v(1_{v^*} \odot e_g \odot 1_{v_*}) : v^* g^* g_* v_* \Rightarrow \emptyset_{\text{cod}(v)}.$$

This is [equation \(7.4\)](#) with its invertible comparisons moved to the other side. It covers every interior seam of a rectangular grid. At an empty horizontal boundary the corresponding equality is [equation \(7.3\)](#). These two identities, the triangles, and [equation \(B.1\)](#) account for all identity, association, and interchange checks in the realization.

APPENDIX C. FINITE TERMS AND SIZE

Proposition C.1. *For a small input signature and small original cell collections, the forest envelope, localization, and global realization have small cell collections. For a class-sized input the same specifications define a finite-term calculus in an encompassing universe; they do not by themselves establish local smallness in the original universe. The same conclusions apply to the path-carrier, distributor, and second completion constructions.*

Proof.

- (1) **Bound the forest collections.** For small data, finite paths and finite subdivisions form sets. Forests are finite products of original cell sets, indexed by a set of subdivisions, hence form a set.
- (2) **Bound the localization.** The collection $S(P)$ is a subcollection of the original cells, so is a set as well. Finite terms in these generators and their formal inverses form a set; the congruence generated by the stated relations is an equivalence relation on it, and its quotient is a set.
- (3) **Bound the global cell collections.** For a fixed vertical arrow f , the binding presentations in $\mathcal{B}(f)$ form a subcollection of a finite product of the original horizontal-arrow and cell sets, and hence a set. Therefore the coproduct and quotient in [equation \(7.8\)](#) are sets. After a local binding presentation is fixed, each global cell collection is identified with the localized hom-set in [equation \(7.5\)](#).
- (4) **Bound the distributor completion.** For the distributor construction on small data, paths and the cells of the carrier equipment form sets. Formal categories and distributors are specified by finitely many carriers and cells satisfying equations, so they form subcollections of sets of such tuples. Distributor multicells are subsets of the corresponding carrier cell sets. The forest and localization argument therefore applies a second time. Restricting on the original atomic monoid objects does not change this conclusion.
- (5) **Treat large data in the larger universe.** For large data the same description uses finite terms labelled in the larger universe. Although every individual cell has a finite description, arbitrarily many intermediate atomic arrows can occur in descriptions with fixed many-sided

endpoints. Finite length alone therefore does not give a small collection in the smaller universe.

- (6) **Identify the fragments with preserved size.** The conservativity theorem does give exactly the original size for the single-output fragment and, after reversal, for the single-input fragment. The second completion similarly preserves exactly the size of the primitive single-output distributor collections, without deducing local smallness of its general many-sided profiles in a smaller universe. $\square$

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